

Per-Stream Terminal Valence Closure:

A Boundary Hypothesis and Framework for Empirical Testing

Stephen Takowsky

Law of Fairness LLC

Publishing Details

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Published in the United States of America.

Publisher: Law of Fairness LLC
Author: Stephen Takowsky
Illustrations: Diego Santos

For permissions, corrections, publication updates, and future edition information: team@lawoffairness.com

Abstract

Could the end of a conscious life reveal a mathematical constraint on the experiences that preceded it? Per-Stream Terminal Valence Closure (PTVC) makes that question a precise, falsifiable candidate boundary hypothesis for affective science, consciousness research, measurement theory and stochastic processes. On its primary finite signed-measure branch, it asks whether each separately routed maximal qualifying conscious stream carries exactly equal positive and negative Jordan masses across its complete strict-pre-endpoint domain, so that the terminal signed residual is zero. The equality is intrastream and nontransferable. This compact boundary condition is compatible with the full diversity of lived experience: lives may differ in gross valence variation, duration, intensity, order, agency, and structure.

$$Z_i^\star := \nu_i\left(\Omega_i\right) = V_i^+ - V_i^- = L_i^{\mathrm{phen}}\left(T_i^\star -\right).$$

$$\mathrm{PTVC}_i \Longleftrightarrow V_i^+ = V_i^- < \infty \Longleftrightarrow Z_i^\star = 0.$$

The boundary condition gains scientific force when representation, neutral origin, exposure, stream ownership, endpoint, observation and recovery rules are fixed before outcome inspection. One sharp consequence concerns endpoint timing. Under a declared reference law, let the endpoint be independent of the entire jointly measurable pre-endpoint ledger path, with a timing density positive almost everywhere on an interval. If exact closure holds at that endpoint almost surely, the ledger must equal zero at almost every eligible time on almost every eligible path. This obstruction gives independent, continuously distributed ending models a precise test. State-coupled timing, first hitting, absorption, endpoint atoms and singular carriers offer alternative models with different testable signatures.

PTVC offers a research program built for prospective evaluation. The first steps are independently calibrated measurements and model recovery. Random-cut comparisons, endpoint-surprise tests, completion-potential transport and concealed-outlet experiments then probe distinct consequences under frozen assumptions. A strong candidate should outperform equally informed rivals on held-out data and invite independent replication. Evidence for one proposed mechanism and evidence for universal terminal equality remain distinct: a precise behavioral signature can favor a route while the primary boundary continues to have its own scientific tests. The aim is cumulative discovery, with each experiment providing an interpretable result and a stronger next question. Together, these investigations ask whether conscious experience obeys a previously unrecognized terminal organizing principle.

Section 14 develops a distinctive empirical proposal: mosquitoes assigned to different adult histories encounter a concealed, randomly designated continuing-life outlet, while a separate test assigns the outcome only after commitment. Different history durations, feeding delays and developmental outcomes create additional prospectively declared contrasts. When the assignment is genuinely uniform and independent of recorded choice, the exact matching baseline is 1/N.

Keywords: PTVC; Law of Fairness; valence; signed measures; terminal boundary condition; latent-variable identification; consciousness; endpoint processes; completion potential; affective science; partial identification; model recovery

Contents

Abstract.....i
1. Scientific question, scope, and research opportunity.....1
A boundary hypothesis with a precise domain.....2
What makes the claim scientific.....3
Scientific foundation and open empirical question.....3
Contribution and novelty.....4
2. Formal object and primary finite-measure claim.....6
A latent ledger requires an exposure model.....8
Bipolar and generalized branches.....9
3. Quantifiers, ownership, routing, and falsification.....10
Stream-level adjudication of the universal claim.....11
Bearer and routing rules are causal, not clerical.....11
Modal claims form a partly ordered family.....12
Target and route are separate axes.....12
Decision status remains typed.....13
4. Representation, neutral origin, and delayed entry.....14
Neutral origin must be calibrated independently.....15
Study entry and stream origin are distinct.....16
Joint calibration: a sharp three-interval example.....17
Factorization, scalarity, and gauge freedom.....18
5. Observation, exactness, and the inference ladder.....19
Target geometry and exactness levels.....20
Inference must preserve the object type.....21
Equivalence requires affirmative evidence.....22
Exact atoms require stable inverse recovery.....22
6. Random endpoints: a sharp structural discriminator.....23
Independent positive-density endpoint proposition.....24
What the discriminator reveals.....25
Empirical use.....25
What the proposition establishes and which branches it opens.....26
Approximate dependence yields a graded pressure test.....26
Information required by endpoint enrichment.....28

7. Reachability, endpoint support, and solvency.....29
A residual must be cancelable by an admissible future.....30
Endpoint support localizes the required structure.....30
Endpoint adjudication is versioned evidence.....32
Endpoint branches are empirical alternatives.....32
8. Optional mechanism routes and path-class constraints.....34
Finite variation restricts exact diffusion bridges.....35
Reserve and endpoint require a tested bridge.....36
Mechanism-localization discipline.....36
Action-mediated route.....37
Exact-differential route.....38
Information retained by a forward realization.....40
9. Completion potential and the transport fingerprint.....42
Completion potential.....43
The overidentification fingerprint.....44
Conjunctive differential signature.....45
Random-cut complementarity.....45
Joint restrictions across a complete partition.....47
Randomized remainder and delayed-treatment localization.....48
10. Rival tournament under equal information budgets.....51
History information versus predictive approximation.....54
11. Existing evidence and the scientific opportunity.....56
Established neighboring results.....57
Plausibility signals and evidential roles.....58
How independent convergence builds confidence.....58
Endpoint, horizon, and consciousness literatures.....59
12. Measurement, missingness, and endpoint validity.....61
Representation before accumulation.....62
Observation coverage completes the estimand.....62
Missing not at random is a central rival.....63
No-lookahead and reactivity.....63
Replicate channels and recoverability.....64
13. Confirmatory decision structure and model recovery.....66
Recovery before interpretation.....68

Negative controls and open-set alternatives....69
Unresolved cases remain visible....69
14. Experimental program: calibration and mosquito terminal-choice tests...71
Stage 0 — protocol fixation, public preregistration, and rigorous design....72
Stage 1 — representation and local complementarity....72
Stage 2 — frozen completion-geometry transport....73
Promotion criterion....73
Immediate analyses in existing data....74
Three high-value experiments....75
Mosquito terminal-choice and future-regime experiment....76
Species, developmental stages, and realistic timing....76
Allocation before hatching and the three histories....78
The twenty-outlet chamber and the five-percent null....80
What the Law-specific prediction requires....84
Timing after feeding or another salient event....86
Hatching and adult-emergence predictions....89
Analysis, repetition, and decisive interpretation....91
15. Endpoint-aligned extension and complete-stream adjudication....94
Observational bridge studies establish the path to terminal tests....95
Prospective endpoint architecture....95
Primary estimands....95
Evidence requirements for complete-stream adjudication....96
Endpoint surprise and promotion....97
16. Exactness, atoms, and symmetric error control....99
Boundary-aware inference....100
Nested-resolution discipline....101
Symmetric error-control budget....101
Sequential evidence across repeated cohorts....102
17. Falsification, safeguards, ethics, and research agenda....104
Results that would narrow or reject a protocol-fixed PTVC claim....105
Known challenges and their research solutions....105
Ethical boundary....107
What should materially change belief....108
Near-term prerequisite and differential predictions....109

Early publishable milestones....109
Cumulative decision bookkeeping....110
Promotion, retirement, and open challenge....110
Ordered research agenda....111
18. Evaluation and reproducibility standard....113
Modular, noncompensatory evaluation....114
Minimum reproducibility capsule....114
Independent audit and discrepancy handling....115
Promotion, retirement, and publication outcomes....115
Declarations....116
References....117
About the Author....132

1. Scientific question, scope, and research opportunity

The scientific question is sharply defined: after independently fixing a qualifying bearer, exhaustive ownership and routing, a complete strict-pre-endpoint truth domain, prospectively fixed valence representation, experienced-exposure rule and clock, neutral origin, and truth-level endpoint, are the positive and negative Jordan masses finite and equal for every separately routed maximal qualifying stream in the declared class?

Prospective-specification convention. Protocol-fixed means written into the study protocol before outcome inspection. Prospectively specified means fixed before outcome inspection without asserting a public record. Publicly preregistered means deposited before outcome inspection in a timestamped external registry with a persistent identifier; cite that identifier when making the claim. Without it, use protocol-fixed or prospectively specified.

A boundary hypothesis with a precise domain

PTVC concerns one cumulative invariant while welfare, opportunity, agency, duration, remembered utility, social outcomes, and gross experience remain distinct dimensions. A stream containing large positive and negative magnitudes may satisfy the same terminal residual condition as a stream containing small magnitudes. This separation preserves the independent importance of rights, justice, suffering prevention, dignity, and practical welfare. Human applications uphold treatment and protection from harm. Animal experiments

receive species-appropriate scientific, welfare, and containment review under the framework in Section 17.

Interpretive scope. PTVC concerns the terminal within-stream relation and leaves path monotonicity and the number of intermediate zero crossings open. Equality of aggregate positive and negative masses does not imply one-to-one event pairing. PTVC does not imply that a bearer knows or computes a ledger, that every urge is corrective, or that psychological motivation is the mechanism. It asserts no debt, punishment, desert, payment, causal payback, or personal guidance.

What makes the claim scientific

Scientific force comes from prospectively fixed gates. The architecture fixes representation, bearer, domain, endpoint, observation, recovery, reference class, and modality before analysis. Route-specific claims compare frozen predictions with equally resourced representation, endpoint-alignment, missingness, instrument-floor, and conditioned-model rivals. Direct target adjudication instead follows its residual and quantifier rules; a verified target-level counterexample need not identify its mechanism. Each passed gate advances only its linked claim.

Scientific foundation and open empirical question

PTVC is mathematically specified and empirically adjudicable under explicit measurement, bridge, and identification assumptions. Existing work on affective measurement,

adaptation, opponent regulation, computational mood, consciousness, terminal trajectories, and missing data supplies candidate instruments, models, and validation methods. This article states a precise target, derives conditional implications, and supplies a research framework designed for prospective preregistration; it reports no new empirical test establishing PTVC. Complete-stream adjudication remains a culminating objective, while a prospectively specified route or consequence restriction can be tested earlier when its implication and observation bridge are independently justified.

Interpretive context. Ideas of balance, restoration, consequence, and completion can generate comparative hypotheses across cultural narratives and lived experience. PTVC-specific scientific weight arises when prospectively specified terminal and cross-module predictions outperform cultural transmission, memory, comfort, cooperation, just-world belief, and other serious rivals.

Contribution and novelty

The article’s distinctive contribution is the typed assembly of signed-measure, stochastic-process, reachability, conditioning, identification, information-theoretic, and model-recovery tools around the PTVC target. It links an exact intrastream terminal condition to random-endpoint, compensator, partition-covariance, retained-state, action-mixture, and transport restrictions. The derivations use established mathematical results under explicitly stated assumptions; they do not present those

results as newly discovered general theorems. Joint-calibration examples, measurement-error identities, and sequential decision rules connect the formal target to assessable studies. Optional mechanisms remain separate from the target, and finite sampling reaches the strict all-qualifying-stream claim ([SET-ALL], defined in Section 3) only through an independent quantifier bridge.

A concrete question for the reader. Imagine a mosquito entering one of many physically equivalent compartments before any subsequent continuing-life assignment is drawn. Independent uniform assignment predicts a precise one-in-N coincidence rate, even if the animal strongly prefers one physical side. A preregistered deviation would challenge that particular independence account; its cause and its relation to the terminal-closure hypothesis would remain separate questions. This example shows the paper’s method: derive a risky observable consequence under stated auxiliary assumptions, expose it to serious rivals, and let each result revise only the claim it actually tests.

2. Formal object and primary finite-measure claim

Fix a separately routed maximal qualifying stream i and its measurable complete strict-pre-endpoint truth domain (Ω_i, Σ_i), and a prospectively declared signed phenomenal representation. On the primary branch, ν_i: $\Sigma_i \to \mathbb{R}$ is a finite signed measure with Jordan decomposition $\nu_i = \nu_i^+ - \nu_i^-$ into mutually singular positive measures. V_i^+ and V_i^- are their finite Jordan masses; their difference is the terminal signed residual.

Primary finite signed-measure branch

$$\nu_i = \nu_i^+ - \nu_i^-, \qquad \nu_i^+ \perp \nu_i^-.$$

$$V_i^{\pm} := \nu_i^{\pm}(\Omega_i), \qquad V_i^{\text{gross}} := V_i^+ + V_i^- = |\nu_i|(\Omega_i).$$

$$Z_i^\star := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^\star -).$$

$$\text{PTVC}_i \iff V_i^+ = V_i^- < \infty \iff Z_i^\star = 0.$$

The claim is exact, intrastream, and nontransferable: a residual in one stream cannot be canceled by experience in another. It is also compatible with nonzero gross variation. The mixed-sign case $V_i^+ = V_i^- > 0$ is therefore structurally distinct from a degenerate stream with no phenomenal exposure. Vector-valued, bipolar, nonadditive, or generalized terminal-neutrality branches remain possible, but they are not silently treated as the scalar additive claim. The universal quantifier is conditional on admissibility of the protocol-fixed finite representation; a broader all-stream claim requires a separate domain-coverage or branch-spanning result.

A latent ledger requires an exposure model

On an absolutely continuous subbranch, a latent ledger can be written as an exposure-weighted integral. The general display retains a left-limit term, but on the primary whole-stream branch S_i^{θ} is the truth-domain origin and $L_i^{\text{phen}}(S_i^{\theta}-)=0$; later entry uses the delayed-entry decomposition on the half-open observed interval [a, b). Here $g_{\theta,i}$ and q_i^{exp} are the two factors in the prospectively frozen exposure-density convention. On the primary convention, $g_{\theta,i}$ is a dimensionless time-chart, unity, or ownership factor and q_i^{exp} has exposure-per-time units; on the unity-clock alternative, $q_i^{\text{exp}}\equiv 1$ and $g_{\theta,i}$ carries exposure-per-time units. Their product is experienced-exposure density; neither factor may duplicate ownership or exposure already encoded by μ_i^{phen}, and v_i^{real} is realized neutral-relative phenomenal value per unit exposure.

Exposure-weighted latent ledger

$$L_i^{\text{phen}}(t) = L_i^{\text{phen}}\,(S_i^{\theta}-) + \int_{S_i^{\theta}}^{t} g_{\theta,i}(s)\, q_i^{\text{exp}}(s)\, v_i^{\text{real}}(s)\, ds.$$

Additivity is the declared representation commitment. Once a signed measure is defined, $\nu_i(\Omega_i)=\nu_i^{+}(\Omega_i)-\nu_i^{-}(\Omega_i)$ is an identity. PTVC adds the substantive boundary restriction that the independently defined residual equals zero at the complete endpoint. Constructed zero-integral functions establish mathematical coherence and help design controls; the empirical opportunity is to determine whether independently calibrated natural phenomenal histories satisfy the same boundary.

Primary target: a boundary property of a complete, independently defined stream, distinguished from normalization rules, population averages, and narrative judgments.

Bipolar and generalized branches

The net signed measure need not exhaust phenomenal structure. On a bipolar branch, let β^+ and β^- be finite nonnegative component measures on the same domain, with $\nu = \beta^+ - \beta^-$. The minimality of the Jordan decomposition gives a unique nonnegative common measure ζ such that

$$\beta^+ = \nu^+ + \zeta, \qquad \beta^- = \nu^- + \zeta.$$

Consequently, primitive gross magnitude and net total variation satisfy

$$\beta^+(\Omega) + \beta^-(\Omega) = |\nu|(\Omega) + 2\zeta(\Omega).$$

The common measure ζ is nonnegative. Adding it to both Jordan components leaves ν unchanged (Folland, 1999). Consequently, net history bounds primitive gross magnitude below but not above without an overlap restriction. Equal net histories can conceal different simultaneous component magnitudes; bipolar phenomenology is not assumed. Generalized Terminal Neutrality (GTN) remains a separate nonadditive or functional-zero branch unless a valid bridge connects it to the additive or bipolar terminal object.

3. Quantifiers, ownership, routing, and falsification

Stream-level adjudication of the universal claim

The primary strict-universal formulation is [SET-ALL]: every separately routed maximal qualifying stream in the protocol-fixed truth-domain class satisfies exact closure. A single gate-valid stream whose nonempty truth-containing residual set (an assumption-guaranteed outer set containing the latent residual) excludes zero is a deterministic counterexample to that protocol-fixed universal branch. A frequentist confidence set supports only its declared coverage statement; a posterior probability is conditional on its protocol-fixed prior and model; and a point estimator remains subject to its uncertainty analysis. Population means, favorable subgroups, or transfers between streams cannot average away a verified counterexample. Probabilistic branches retain their declared probability and error guarantees.

Bearer and routing rules are causal, not clerical

The unit of analysis is a maximal qualifying conscious stream under a prospectively fixed bearer and ownership rule. Splitting one stream after observing its residual, merging streams, dropping weakly observed intervals, or duplicating an event can create apparent balance. Routing must be exhaustive, nonduplicative, time-consistent, and outcome-independent. If individuation is partially identified, report the set over all admissible routings. Unresolved positive and negative contributions enter separate nonnegative ownership-slack accounts and may not cancel each other. Population claims

additionally require declared stream-level inclusion probabilities or survey weights; convenient observed streams do not define the target population. The protocol is fixed before quantifying over streams, histories, continuations, or policies.

Modal claims form a partly ordered family

Existence of a completing continuation, positive-probability completion, fixed-policy probability-one completion, almost-sure attainability, limit-sure attainability, topological-support containment, and closure of every admissible continuation are distinct. Almost-sure attainability requires a single admissible policy attaining probability one; limit-sure attainability permits success probabilities arbitrarily close to one without requiring an attaining policy (Asadi et al., 2025; Chatterjee et al., 2016). Executing a policy additionally requires identifying or constructing it. Every experiment fixes its modality, policy or intervention, continuation class, and evidence object. Realized-policy, all-policy, and all-intervention closure impose different scopes; the latter two require a structural model and an admissible counterfactual domain. No fixed-policy result establishes those stronger scopes or [SET-ALL] without a separate bridge.

Target and route are separate axes

Target support concerns the terminal residual under a validated bridge. Route support concerns an optional mechanism proposed to generate or constrain closure. Their separation strengthens

inference: route failure leaves the target open, while route fit advances the target verdict only through a proved entailment.

Evidence remains branch-matched: exact-zero, finite-band, probability, and mechanism claims each use their corresponding support, rejection, and abstention rules.

Decision status remains typed

Public reporting uses a four-axis result vector (TargetStatus, RouteStatus, ComparatorStatus, and InferenceStatus), together with a supporting record of prerequisites, target specification, recovery, and attainment. The confirmatory reporting discussion in Section 13 fixes the controlled vocabularies used in this article, and each axis remains independently typed. No composite score may override an adverse target result, rescue a failed prerequisite, conflate target geometries or modalities, or turn route superiority into proof of the terminal law.

4. Representation, neutral origin, and delayed entry

Neutral origin must be calibrated independently

A scalar ledger requires a sign convention, cross-sign comparability, an exposure unit, and a neutral origin. These are measurement hypotheses. Under the admissible affine recoding v′=ai+biv with bi>0, a stream-constant positive scale preserves a zero residual, whereas on positive-exposure records an additive shift preserves zero only when ai=0. The origin must therefore be calibrated on nonterminal data or bounded as an uncertainty set before the terminal residual is inspected. The admissible origin class (basis, complexity, envelope, and calibration sample) must be frozen. If it can select a record-specific function that forces zero, the origin is unidentified and the confirmatory result is No adequate estimate (NAE).

Origin sensitivity under affine transformation

$$\tilde{Z}_i^{\star} = a_i E_i^{\mathrm{tot}} + b_i Z_i^{\star}, \qquad b_i > 0, \qquad E_i^{\mathrm{tot}} = \mu_i^{\mathrm{phen}}(\Omega_i) < \infty.$$

If total exposure is positive, choosing $a_i = -b_i Z_i^{\star} / E_i^{\mathrm{tot}}$ forces the transformed residual to zero. That operation is precisely why completed-record mean-centering is prohibited as confirmatory evidence. A valid cross-sign program perturbs independently validated positive and negative channels, estimates their exchange map, tests state dependence and hysteresis, and freezes the admissible origin set before endpoint fitting.

Study entry and stream origin are distinct

Because most studies start late and stop before the true endpoint, whole-stream claims require one joint truth-containing prefix–observed–suffix object on the same latent scale. It must be narrow enough to decide the protocol-fixed exact or finite-band target; separate component bounds become decisive only through that joint object.

Delayed-entry decomposition

$$Z_{i,\mathrm{pre}}(a) := L_i^{\mathrm{phen}}(a-),$$

$$Z_{i,\mathrm{obs}}([a,b)) := L_i^{\mathrm{phen}}(b-) - L_i^{\mathrm{phen}}(a-),$$

$$Z_{i,\mathrm{suf}}(b) := L_i^{\mathrm{phen}}(T_i^\star-) - L_i^{\mathrm{phen}}(b-),$$

$$Z_i^\star = Z_{i,\mathrm{pre}}(a) + Z_{i,\mathrm{obs}}([a,b)) + Z_{i,\mathrm{suf}}(b).$$

The three terms must be constructed jointly so their dependence and simultaneous guarantee are preserved. Unbounded components do not automatically make their sum uninformative: dependence can restrict the joint projection, and even an unbounded residual set can exclude zero. A nonempty valid outer set contained in $[1, \infty)$, for example, is adverse to exact closure under its stated deterministic or statistical guarantee. If a valid set intersects both the protocol-fixed acceptance region and its complement, the result is indeterminate. No adequate estimate (NAE) applies when no valid target-linked decision object can be constructed, not merely when an interval is unbounded. An empty compatibility

set diagnoses inconsistency of the maintained model and data restrictions; it is not by itself a verified terminal residual. A measured branch keeps the same accounting on its declared scale without silently promoting it to the phenomenal ledger.

Identification gate: a stable sign, origin, unit, and cross-sign scalarization promotes the scalar branch; if one component remains unresolved, the vector-valued alternatives stay available for corresponding tests.

Joint calibration: a sharp three-interval example

A constructed example shows why shared unknowns must remain coupled. For three signed interval contributions, suppose the jointly valid restrictions are

$$x_1 + x_2 \in [0.7, 0.9], \qquad x_2 + x_3 \in [0.3, 0.5],$$

$$x_2 \in [-0.2, 0.2].$$

Writing $J = x_1 + x_2 + x_3$ gives

$$J = (x_1 + x_2) + (x_2 + x_3) - x_2 \in [0.8, 1.6].$$

The feasible vectors $(0.5, 0.2, 0.1)$ and $(1.1, -0.2, 0.7)$ attain the two endpoints, so the range is sharp. Without the bound on x_2, the two observed sums can remain fixed while J becomes arbitrarily positive or negative. An additional measurement of the shared component can therefore identify the total when repeated measurements of the two sums cannot. These are synthetic units, not participant data. Statistical restrictions must hold jointly with the claimed coverage;

projecting separate marginal intervals without simultaneous error control does not inherit that guarantee.

Factorization, scalarity, and gauge freedom

On a density branch, phenomenal contribution can be factorized into exposure and neutral-relative value per unit exposure. Positive predictable rescaling of one factor with inverse rescaling of the other leaves the signed measure unchanged, so exposure and intensity are not separately identified by their product. Clock time, subjective duration, attention, consciousness density, and unity weights must be compared under independent manipulations rather than selected for terminal fit. Raw ratings, ranks, embeddings, factor scores, and logits are not automatically cardinal. Additive-conjoint independence or separability, cancellation, solvability, and held-out consistency test whether one cross-sign scalar is defensible (Luce & Tukey, 1964); bipolar co-occurrence tests whether positive and negative components must instead coexist (Cacioppo & Berntson, 1994; Larsen et al., 2001). An adequate reproducible violation fails the scalar prerequisite; nonidentification or invalidity yields NAE. Prespecified vector, ordered, set-valued, and GTN branches remain open. That representation result is not adverse TargetStatus evidence for every branch.

5. Observation, exactness, and the inference ladder

Let P_i^Z denote the latent terminal-residual law on the declared truth space. Observed reports, physiology, behavior, neural signals, and clinical records arise through a Markov kernel $K_{i,\eta}^{obs}$ from that latent space to a specified observation space; η indexes instrument, resolution, care context, communication, sedation, cadence, retention, and other declared conditions. The observed law $P_{i,\eta}^{obs}$ is the kernel push-forward of P_i^Z. Observed variance contraction therefore need not be latent contraction. For longitudinal observations, this kernel also incorporates the declared conditional law of latent histories given the terminal residual. That history law must be specified and validated, or the latent state enlarged; a terminal-residual distribution alone does not determine trajectory observations.

Observation-kernel push-forward

$$P_{i,\eta}^{\mathrm{obs}}(A)=\int K_{i,\eta}^{\mathrm{obs}}(A \mid z)\ P_i^Z(dz).$$

Target geometry and exactness levels

For the latent exact target, a nonempty truth-containing residual set is exact-compatible only when it equals {0}. It is exact-adverse when it excludes zero, and otherwise is exact-indeterminate. A confidence set containing zero is not a deterministic truth-containing singleton.

For a finite-band target, the scientific acceptance set is fixed in the protocol. The residual set is band-equivalent when wholly inside it, band-adverse when disjoint, and band-indeterminate when it overlaps both acceptance and

nonacceptance. The interval $[-\varepsilon, \varepsilon]$ is a scalar special case; a law-level band-probability claim separately specifies its stream or population law, minimum mass threshold, and error rule.

An exact zero atom means that the latent terminal law assigns positive mass to zero. Shrinking observed bands cannot establish such an atom without validated inverse recovery, deconvolution, or partial-identification assumptions.

Law-level quantifiers distinguish probability-one closure under a declared random-stream law from set-universal closure over every qualifying stream. Neither follows from a finite sample of compatible streams or merely from a large estimated atom mass.

Inference must preserve the object type

Scientific acceptance geometry and epistemic uncertainty are different objects. The acceptance set is fixed prospectively; uncertainty widens the identified region, truth-containing outer set, confidence set, credible set, or sensitivity region rather than that acceptance set (Manski, 2003). A frequentist confidence set is a random set whose protocol-fixed procedure controls repeated-sampling coverage; it is not a deterministic truth-containing set. A Bayesian decision remains prior- and model-conditional, while a deterministic falsifier requires a truth-containing set. Origin, endpoint, routing, missingness, kernel, and numerical uncertainty must be propagated. No geometric relation silently converts finite-band evidence into exact or universal closure.

Equivalence requires affirmative evidence

Equivalence is supported through prospectively specified margins, family-wise error control, boundary-aware asymptotics, mixture alternatives, and model-recovery performance rather than through nonsignificance alone (Schuirmann, 1987; Lakens, 2017; Self & Liang, 1987). A nonidentifying observation map keeps inference at the measured scale until a valid latent bridge is supplied.

Exact atoms require stable inverse recovery

An injective observation map on a protocol-fixed law class does not by itself guarantee stable finite-sample recovery of mass at exactly zero. Diffuse laws concentrated arbitrarily near zero can resemble an atom after noise, rounding, censoring, smoothing, or kernel uncertainty. Recovery studies must therefore include true atoms, near-zero diffuse alternatives, locally discrete laws without a zero atom, mixtures, singular components, quantized channels, observation floors, and perturbed kernels. If the pipeline cannot distinguish those generators at the protocol-fixed resolution, the proper conclusion is a finite band, partial identification, or No adequate estimate. Under a locally smoothing observation kernel, atomless latent laws can concentrate arbitrarily near zero and become arbitrarily close in observed total variation to a true zero atom at every fixed sample size; exactness therefore requires a protocol-fixed separation or structural restriction, not merely more observations under the same unresolved kernel.

6. Random endpoints: a sharp structural discriminator

Exact closure at an independently timed endpoint with a density positive Lebesgue-almost-everywhere imposes a strong and informative terminal signature. Under the assumptions below, for almost every eligible path under the declared conditional reference law, the pre-endpoint ledger equals zero for Lebesgue-almost every time on the interval.

Independent positive-density endpoint proposition

Let E_i be a prospectively declared positive-probability event on which the stream, frozen routing and truth domain, finite ledger, and pre-killing extension exist on a deterministic interval I; write $P_i^0=P(\cdot|E_i)$. Under this same law, let $\tilde{L}_i(t-)$ be jointly measurable, $\mathscr{F}_i^L=\sigma\{\tilde{L}_i(s-):s\in I\}$, and T_i^{exo} independent of $\mathscr{F}_i^L$. Assume T_i^{exo} is I-valued, has a Lebesgue density positive almost everywhere on I, and $P_i^0\{\tilde{L}_i(T_i^{exo}-)=0\}=1$.

Independent positive-density endpoint obstruction

$$0 = P_i^0\left(\widetilde{L}_i\left(T_i^{\mathrm{exo}}-\right)\neq 0\right) = \int_I P_i^0\left(\widetilde{L}_i\left(t-\right)\neq 0\right) f_{T_i^{\mathrm{exo}}}(t)\,dt.$$

The integrand is nonnegative. Tonelli’s theorem and the positive density give the product-measure statement: $P_i^0\{\tilde{L}_i(t-)\neq 0\}=0$ for Lebesgue-almost every $t\in I$; Fubini’s section conclusion equivalently gives $\tilde{L}_i(t-)=0$ for Lebesgue-almost every t on a P_i^0-probability-one set of paths. If paths are left-continuous P_i^0-almost surely, equality holds at every interior time; the left endpoint also requires right continuity or a separate value condition. Thus, on the independent positive-density

branch, exact closure requires the zero-almost-everywhere structure identified by the proposition.

What the discriminator reveals

The proposition identifies the structure a nondegenerate implementation must supply. Candidate implementations include state-coupled endpoint timing, first-hitting or absorbing boundaries, endpoint atoms or singular support, phenomenally active endpoints, routing change, and failure of the pre-killing extension. Each defines a testable branch with distinctive predictions.

Empirical use

Unexpected and weakly anticipated endpoints are high-pressure tests. The analysis must measure residual dependence between the path and endpoint after prospectively declared covariates, report endpoint surprise, and compare state-coupled hazards, selection, observation shutdown, and random-endpoint bridge rivals. An unplanned endpoint is not thereby exogenous; the protocol-fixed dependence, surprise, and rival checks provide a graded sensitivity analysis, while the independent-endpoint proposition applies only when its independence premise is separately justified.

What the proposition establishes and which branches it opens

The branch-specific discriminator prospectively localizes routes. Every nontrivial exact-closure model must show structure beyond independent positive-density timing; the candidates above make that testable. Premises remain fixed before residual inspection. Approximate independence gives graded pressure when endpoints sample nonzero-ledger times. Ledger-dependent timing identifies a route for direct evaluation; the terminal test adjudicates closure.

Approximate dependence yields a graded pressure test

Fix a nonnegative tolerance and a jointly measurable pre-endpoint ledger path. Let the covariates be fixed prospectively, and choose a Borel subset of the analysis interval with finite, strictly positive Lebesgue measure. All subsequent quantities are defined under the declared eligible-stream law. Set

$$X_i := (\widetilde{L}_i(t-))_{t\in I}, \qquad \varepsilon \geq 0, \quad J \subseteq I, \quad 0 < \lambda(J) < \infty.$$

$$g_{i,\varepsilon}\,(t,c) := P_i^0 \left(|\widetilde{L}_i\,(t-)\,| > \varepsilon \mid C_i = c \right)$$

$$p_{i,\varepsilon} := P_i^0 \left(|\widetilde{L}_i \left(T_i^{\text{probe}} - \right)| > \varepsilon \right)$$

$$\rho_i := E_{P_i^0} \left[d_{\text{TV}} \left(P_{X_i, T_i^{\text{probe}} | C_i}, P_{X_i | C_i} \otimes P_{T_i^{\text{probe}} | C_i} \right) \right]$$

Evaluate the conditional kernels at the prospectively declared covariates. The total-variation distance is defined by

$$\mathrm{d_{TV}}(P,Q) := \sup_{A} |P(A) - Q(A)|$$

Assume standard-Borel path and covariate spaces, measurable evaluation, and the required regular conditional laws. Suppose that

$$f_{T_i^{\mathrm{probe}}|C_i}(t \mid C_i) \geq m_J > 0$$

The density bound holds for Lebesgue-almost every time in the declared set and almost surely under the eligible-stream reference law.

Approximate endpoint-dependence occupation bound

$$E_{P_i^0}\left[\int_J g_{i,\varepsilon}\,(t,C_i)\;dt\right] \leq \frac{p_{i,\varepsilon} + \rho_i}{m_J}, \qquad m_J > 0.$$

The total-variation event bound changes the probe off-band event by at most ρ_i; the density floor then gives the display. Consequently, substantial off-band occupation together with rare probe off-band events requires correspondingly large endpoint–ledger dependence after the declared covariate adjustment. Its scientific role is to quantify sensitivity to dependence; independence and SET-ALL remain separate protocol-fixed analyses.

Information required by endpoint enrichment

The same comparison has an information-theoretic form. Under one declared eligible-stream law, fix standard-Borel variables, a ledger path on a common evaluation domain, and covariates before endpoint comparison. For each covariate value, compare the conditional joint law with the product of its conditional marginals. The same measurable tolerance event applies to both laws, including product-law pairs. The conditional event probabilities and their averages are defined below. All logarithms are natural.

$$P_c := \mathrm{Law}((T, \mathcal{L}) \mid C = c), \quad Q_c := P_{T|c} \otimes P_{\mathcal{L}|c},$$
$$A_\varepsilon := \{|L(T-)| \le \varepsilon\}, \quad p_c := P_c(A_\varepsilon), \quad q_c := Q_c(A_\varepsilon),$$
$$p := \mathbb{E} p_C, \qquad q := \mathbb{E} q_C.$$

$$I(T; \mathcal{L} \mid C) \ge \mathbb{E}\, \mathrm{d}_{\mathrm{Ber}}(p_C \| q_C) \ge \mathrm{d}_{\mathrm{Ber}}(p \| q),$$

$$\mathrm{d}_{\mathrm{Ber}}(p \| q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

Relative entropy contracts under event recording, and joint convexity gives the second inequality (Cover & Thomas, 2006). Standard extended-value conventions apply: for p = 1 and 0 < q < 1 the lower bound is log(1/q), while for q = 0 < p it is infinite. Thus enrichment requires endpoint–history dependence, which ordinary state-dependent stopping can also produce. The bound does not identify a unique mechanism or directly measure consciousness.

7. Reachability, endpoint support, and solvency

A residual must be cancelable by an admissible future

For a realized stream-owned prefix h_t and prospective endpoint u, define the admissible continuation class $C_i(t,u|h_t)$ without terminal-outcome leakage. Let Cap_i^+ be the supremum attainable future positive contribution and Cap_i^- the magnitude of the infimum attainable future negative contribution. Any continuation that closes the ledger must contribute the negative of its current value, yielding the necessary directional solvency screen:

Directional reachability necessity

$$-\mathrm{Cap}_i^+(t,u \mid h_t) \le L_i^{\mathrm{phen}}(t) \le \mathrm{Cap}_i^-(t,u \mid h_t).$$

A residual outside this interval makes exact completion at u impossible. An empty continuation class also makes completion impossible, and no capacity supremum is evaluated. An undefined, invalid, or unrecoverable class or decision object yields unavailable or NAE. Passing the interval screen is necessary, not sufficient: the required increment must belong to the actual attainable set, which may have gaps. Compactness can secure attainment of extrema without filling those gaps; completion needs a separate existence or intermediate-value argument under the declared dynamics (Aubin, 1991).

Endpoint support localizes the required structure

If endpoints can occur over broad support, the residual must remain solvent there. Suppose the endpoint counting process has

a predictable, locally integrable intensity on the at-risk set that generates an absolutely continuous compensator of finite mass on the analysis interval; assume no simultaneous ledger–endpoint jump. Intensity weighting cannot create a new exact atom at zero from an occupation law that is atomless there. Exact atomic closure instead requires pre-existing atomic occupation, absorption or stickiness, a first-hitting construction, a predictable endpoint atom, a singular compensator, or a justified limiting mechanism. Atomic, singular-compensator, first-hitting, and limiting branches remain separate.

For a precise form, let $T>0$ be a stopping-time endpoint, possibly infinite, and let $N_t=\mathbf{1}_{\{T\le t\}}$ have predictable compensator A_t . Assume that $L(t^-)$ is predictable in the same filtration and defined on the needed at-risk domain. For a finite horizon u and $\delta\ge 0$, compensation gives

$$\Pr\{T\le u,\ |L(T^-)|>\delta\}$$

$$=\mathbb{E}\int_{(0,u]}\mathbf{1}_{\{|L(t^-)|>\delta\}}\,dA_t.$$

The integral against dN_t is the indicator of the displayed endpoint event; replacing dN_t by its compensator inside expectation proves the identity (Andersen et al., 1993; Protter, 2005). If exact incoming closure holds, the right side at $\delta=0$ is zero, and its nonnegative integral vanishes almost surely. For $dA_t=Y_t\lambda_t\,dt$, with $Y_t=\mathbf{1}_{\{T\ge t\}}$, compensator mass is confined to the incoming zero set. If that set has zero at-risk time occupation and the compensator is absolutely continuous

with finite mass, no endpoint can occur by u with positive probability. Singular or atomic endpoint laws require their own accounting. Projecting to an observed filtration does not justify multiplying a projected hazard by a separately projected ledger-band probability; their joint observation model must be derived.

Endpoint adjudication is versioned evidence

Candidate crossing, Estimated endpoint, Confirmed under protocol, Revised endpoint, or No adequate estimate are distinct statuses. Dwell time, return logic, competing events, reliability, and revision rules are frozen before ledger inspection. Later recovery or reclassification preserves the original analysis version and propagates endpoint uncertainty; the last available observation is never forced to be the truth-level endpoint. Reachability measures future capacity; actual selection, mechanism, and completed closure are evaluated through their corresponding protocol-fixed tests.

Endpoint branches are empirical alternatives

Independent positive-density timing predicts the obstruction above. First hitting or absorption can place endpoint mass on a zero surface with a singular compensator rather than an ordinary hazard. Predictable atoms allow mass at discrete boundaries; singular carriers require carrier-specific occupation tests. Endpoint-inclusive branches require supported phenomenal mass and ownership at the boundary. Biological death remains distinct from the truth endpoint. The series separately proposes a

biological-death admissibility condition: biological death may not occur before the relevant stream satisfies its declared closure condition. This condition is additional to primary PTVC and does not require death when closure is reached. A stream may permanently end after closure while nonconscious biological processes continue, including under artificial life support. Any death-associated contribution follows the declared strict-pre-endpoint or inclusive convention. Routing changes require prospective junction rules and nonduplication. Each branch needs its own evidence; defining an endpoint as a zero hit makes closure true by construction, whereas predicting coincidence with an independently adjudicated endpoint permits a route test.

Unexpected or weakly anticipated natural endpoints are especially informative because a local controller may lose sufficient actionable horizon while a global or state-coupled route may transport. Human endpoint research remains observational and preserves survival and care. The separately specified mosquito proposal in Section 14 experimentally varies terminal contingencies and their timing; its endpoint, observation, and biological-death assumptions are specified independently.

8. Optional mechanism routes and path-class constraints

PTVC is deliberately mechanism-neutral, allowing reduced reserve-coupled routes, conditioned bridges, absorbing processes, constrained controllers, and other constructions to compete through their distinctive assumptions and predictions. The core target remains independently evaluable across those routes.

Finite variation restricts exact diffusion bridges

A finite signed-measure ledger has finite variation; a continuous version therefore has zero continuous quadratic variation. Suppose an exact path bridge identifies it with a reserve-coupled semimartingale whose continuous local-martingale coefficient is the product of a positive ledger–reserve conversion constant, the remaining reserve, and the diffusion coefficient, while the remaining terms have finite variation. Writing continuous quadratic variation in the displays below, exact path equality requires both increments to vanish:

Finite-variation obstruction to an exact diffusion bridge

$$\left[L_i^{\mathrm{phen}}\right]_t^c - \left[L_i^{\mathrm{phen}}\right]_{S_i^\theta}^c = 0.$$

$$[L_i^{\mathrm{rc}}]_t^c - [L_i^{\mathrm{rc}}]_{S_i^\theta}^c = c_{\mathrm{LU}}^2 \int_{S_i^\theta}^t U_i(s)^2 \sigma_i(s)^2 \, ds = 0.$$

Here the product of route-specific reserve and diffusion coefficient is assumed locally square-integrable in time, almost surely, so the stated stochastic integral is defined. Exact path equality then forces their product to vanish for Lebesgue-almost

every time, almost surely; wherever reserve remains positive, the diffusion coefficient must vanish in the same sense. A nondegenerate diffusion may remain an approximate, terminal-only, or consequence bridge, but not an exact continuous finite-variation path identity without resolving this mismatch (Protter, 2005). Jumps require separate quadratic-variation accounting.

Reserve and endpoint require a tested bridge

Mechanism-level reserve inference begins with a proxy distinct from ordinary time-to-endpoint prediction. A credible proxy is constructed without terminal-neutrality labels, adds predictive value beyond prognosis, frailty, disease severity, treatment, care setting, device dropout, and observation availability, and transports across sites and perturbations. A separately tested coincidence bridge can then connect route-state exhaustion with the truth-level stream endpoint.

Mechanism-localization discipline

An intervention may change assignment, information, policy, action, physical transition, outcome, retention, endpoint, routing, or observation. The earliest distinguishable change in a conditional law is a candidate causal locus only under the declared identification, timing, measurement-lag, and observation assumptions. In continuous time, causal weighting additionally requires positivity and an appropriate local-independence structure (Røysland, 2011). Downstream outcomes of one alteration must be analyzed with their

dependence retained, rather than counted as independent confirmations.

Action-mediated route

Under the protocol-fixed action-first factorization, let h be the predecision history, g the declared continuation regime, and $\mathcal{A}^{\mathrm{do}}(h)$ the measurable admissible action domain. Let π be the fixed policy and p^{comp} (a | h, g) the measurable probability that the complete frozen continuation closes after action a under that regime. F_{comp} denotes the induced joint action–suffix completion event. Assume the policy assigns probability one to $\mathcal{A}^{\mathrm{do}}(h)$. Fixed-policy probability-one completion then holds exactly when p^{comp} equals one for policy-almost every selected action. This is a policy-mass statement; topological-support containment, motivation, and endpoint support require separate assumptions.

$$p^{\mathrm{comp}}(a \mid h,\ g) \ :=\ P_g\left(F_{\mathrm{comp}} \mid h,\ a\right)$$

$$\pi\left(\mathrm{A}^{\mathrm{do}}(h) \mid h,g\right) = 1,$$

$$P^{\pi,g}\left(F_{\mathrm{comp}} \mid h\right) = \int_{\mathrm{A}^{\mathrm{do}}(h)} p^{\mathrm{comp}}(a \mid h,g)\ \pi\left(da \mid h,g\right),$$

$$P^{\pi,g}\left(F_{\mathrm{comp}} \mid h\right) = 1$$

$$\iff\ p^{\mathrm{comp}}(a \mid h,g) = 1 \quad \pi(\cdot \mid h,g)\text{-almost everywhere.}$$

Finite soft guidance is weaker than exact selection. Write $q(a)=p^{\mathrm{comp}}(a\mid h,g)$ and fix a baseline policy π_0, a finite exponent $\beta>0$, and $Z_\beta=\int q(a)^\beta\,\pi_0(da)>0$. Reweighting by q^β yields

$$\pi_\beta(da)=Z_\beta^{-1}q(a)^\beta\pi_0(da),\qquad P^{\pi_\beta,g}(F_{\mathrm{comp}}\mid h)=\frac{Z_{\beta+1}}{Z_\beta}.$$

The ratio equals one exactly when π_0 assigns no mass to $0<q<1$; positivity of Z_β then requires mass at $q=1$. This follows by integrating $q^\beta(1-q)$. Any retained intermediate-completion mass prevents finite probability-one completion. A sequence approaching one need not attain one. Under partial observation, a robust guarantee also requires an admissible action common to the probability-one action sets of all compatible histories, with measurable selection and time-consistent continuation. A high posterior-average value is a different guarantee. These distinctions make a concealed-outlet selector in Section 14 an explicit information and policy hypothesis rather than a consequence of the terminal equation alone.

Exact-differential route

A separate route treats the ledger as a path integral of a one-form on a sufficiently rich state space. If the one-form is the differential of a potential Φ on the relevant domain, an open path from S to T has an endpoint-dependent integral, whereas every admissible closed loop has zero integral. Hysteresis, nonzero

circulation, or path-dependent residue under frozen coordinates rejects that exact route unless the declared state is enlarged and independently revalidated.

Exact-differential route and loop diagnostic

$$\omega = d\boldsymbol{\Phi} \quad \Rightarrow \quad \int_{\gamma_{S\to T}} \omega = \boldsymbol{\Phi}\left(x_T\right) - \boldsymbol{\Phi}\left(x_S\right).$$

$$\oint_{\Gamma} \omega = 0.$$

Empirical weight comes from held-out route predictions beyond generic mathematical existence, including the generic ability to condition an ordinary process on completion; the attainability distinctions in Section 3 remain in force.

Information retained by a forward realization

A nominated causal state can be tested by what it retains about the prefix. At one fixed admissible cut, let L be the accrued contribution and R the entire remainder under the same representation, domain, routing, and endpoint rules. Let X include all state information used to generate the continuation, including its endpoint process. Suppose L and R are conditionally independent given X and have finite second moments. Put $D = \mathbb{E}[(L+R)^2]$. Conditional variance decomposition gives

$$D = \mathbb{E}\,\mathrm{Var}(L \mid X) + \mathbb{E}\,\mathrm{Var}(R \mid X)$$

$$+\mathbb{E}[(\mathbb{E}[L \mid X] + \mathbb{E}[R \mid X])^2]$$

$$\geq \mathbb{E}\,\mathrm{Var}(L \mid X).$$

If $R = -L$ almost surely, X must determine L up to null events. With real L and standard-Borel X, the exact conclusion needs no moment assumption: conditional independence and $R = -L$ make each conditional rational-threshold probability equal to its square, forcing the conditional distribution of L to be a point mass. Approximate closure instead has the displayed error floor. The required information may be bodily, molecular, neural, or distributed; it need not be consciously recalled. A noisy assay of X can lose information retained by the actual state. The candidate state must also be defined before testing; appending the measured prefix after the fact establishes sufficiency by construction, not a mechanism.

This tests the stated screening-off realization, not every formulation of PTVC.

For a Gaussian L with variance $s^2 > 0$, the same screening condition and $0 < D < s^2$ give a quantitative bound in bits:

$$I_2(L;X) \geq I_2(L;-R) \geq \tfrac{1}{2}\log_2 \tfrac{s^2}{D}.$$

Here $-R$ reconstructs L with squared error D. The first step is data processing for $L \longrightarrow X \longrightarrow R$; the second is the Gaussian squared-error bound (Cover & Thomas, 2006). A state with at most K values must satisfy $D \geq s^2/K^2$. The bound is trivial when $D \geq s^2$ and excludes finite K at zero distortion. No recentering of the prefix is required.

9. Completion potential and the transport fingerprint

A discriminating near-term route is a prospectively frozen completion potential transported across independent modules.

Completion potential

Let $H_{i,t}$ be the frozen pre-endpoint filtration completed under P_i^0, with $\bar{\Xi}_{i,t}$ and the current-risk indicator $\mathbf{1}_{\{T_i^\star > t\}}$ measurable with respect to $H_{i,t}$. For positive-probability $A_{i,\varepsilon}$, require on $\{T_i^\star > t\}$ that the following equality hold almost surely:

$$P_i^0(A_{i,\varepsilon} \mid H_{i,t}, T_i^\star > t) = P_i^0(A_{i,\varepsilon} \mid \bar{\Xi}_{i,t}, T_i^\star > t) = h_{i,\varepsilon}(t, \bar{\Xi}_{i,t})$$

Otherwise augment the state or retain history. Nested bands require potentials monotone in ε. The shrinking-band limit is an exact-atom statement only under the stated regular-conditional construction.

Positive-probability completion potential

$$A_{i,\varepsilon} := \{|Z_i^\star| \le \varepsilon\}.$$

$$h_{i,\varepsilon}(t,\xi) := P_i^0(A_{i,\varepsilon} \mid \bar{\Xi}_{i,t} = \xi, T_i^\star > t).$$

$$\lim_{\varepsilon \downarrow 0} h_{i,\varepsilon}(t,\xi) = P_i^0(Z_i^\star = 0 \mid \bar{\Xi}_{i,t} = \xi, T_i^\star > t).$$

Let $\mathbb{E}_i^0$ denote expectation under P_i^0. Require $P_i^0(A_{i,\varepsilon} \mid H_{i,0}) > 0$ almost surely on the declared risk set. The resulting process is a generic nonnegative normalized conditional-probability martingale and supplies calibration checks. PTVC-specific route evidence arises when protocol-

fixed held-out relations transport beyond this generic behavior. Its declared domain requires the stated positive-probability condition; probability-zero conditioning uses a declared regular-conditional bridge, density limit, or other justified construction (Doob, 1957; Chetrite & Touchette, 2015).

Normalized completion-event martingale

$$M_{i,\varepsilon}(t) := \frac{P_i^0\left(A_{i,\varepsilon} \mid \mathrm{H}_{i,t}\right)}{P_i^0\left(A_{i,\varepsilon} \mid \mathrm{H}_{i,0}\right)}.$$

$$E_i^0\left(M_{i,\varepsilon}(t) \mid \mathrm{H}_{i,s}\right) = M_{i,\varepsilon}(s), \qquad 0 \le s \le t.$$

The overidentification fingerprint

Estimate one potential on training modules without test-endpoint leakage; freeze its state, normalization, tolerance family, support, and observation pipeline; then predict held-out transition dynamics, action odds, endpoint support, terminal-band behavior, and resolution transport. Observational completion potential under P_i^0 and interventional completion potential under a declared policy are different estimands and may not be exchanged without identification and transport assumptions.

A Doob transform provides a generic conditioned-law benchmark; held-out transport beyond adequate conditioned rivals supplies PTVC-specific route support.

Conjunctive differential signature

Specificity comes from a conjunctive pattern of independently tested signatures. The primary proposed conjunctive route package requires reserve-dispersion behavior; two-time covariance and perturbation response beyond bridge rivals; endpoint-surprise calibration; signed-dose compensation with assigned, delivered, experienced, and phenomenal dose kept distinct; residual incremental value; residual compression sufficiency; transport across transition, action, endpoint, terminal-support, resolution, and observation modules; and equal-budget rival failure on common support. Incremental value and compression remain separate claims.

Random-cut complementarity

At a prospectively randomized cut, finite additivity links the accrued prefix and future remainder to the complete residual. The fixed disjoint half-open truth-domain slices assign any cut-time atom once. The identity is bookkeeping; PTVC adds an independently tested zero restriction. The observed remainder must come from post-cut data or a validated terminal-remainder bridge, never subtraction of the measured prefix from the tested terminal target. Use a two-stage within-stream design with randomized cuts, independent prefix and remainder calibration, stream-specific intercepts, cluster-aware uncertainty, and the protocol-fixed errors-in-variables model below. In the display, u_{ik} and v_{ik} are the protocol-fixed prefix and remainder observation or bridge errors; their joint law across cuts and

channels is prospectively specified. Estimate β freely: $\beta = 1$ is the additivity gate, equivalently a protocol-fixed prefix coefficient of -1. On the exact direct-partition branch, $\beta = 1$, and $\alpha_i = Z_i^\star$ follows only when the relevant combined error location is fixed at zero by design or independent calibration. These are structural restrictions, not observed-data identification results. Interpret or test α_i only after a valid errors-in-variables route identifies or bounds the joint error location, scale or covariance, and admissible bias through repeated measurement, validation data, instruments, structural restrictions, or bounded partial identification. Any additional latent residual belongs to a separately protocol-fixed bridge or model-misspecification branch, not the exact ledger identity. A bounded-episode zero intercept is local unless the cut partitions the complete qualifying stream or a protocol-fixed prefix–observed–suffix bridge covers the omitted domain; a pooled zero is a preliminary result that advances through stream-specific adjudication. Repeated cuts are dependent within stream and do not replace stream-specific adjudication. The slope gate tests consistency with the declared additive partition; it does not by itself validate phenomenal additivity. The independently tested terminal restriction supplies the corresponding PTVC test.

Random-cut additivity and stream-specific residual adjudication

$$Z_i^\star = L_{ik}^{\mathrm{pre}} + L_{ik}^{\mathrm{post}}.$$

$$\widehat{L}_{ik}^{\mathrm{pre}} = L_{ik}^{\mathrm{pre}} + u_{ik},$$

$$\widehat{L}_{ik}^{\mathrm{post}} = \alpha_i - \beta L_{ik}^{\mathrm{pre}} + v_{ik}.$$

Joint restrictions across a complete partition

The random-cut relation extends to a fixed finite partition of a complete stream domain. Under one prospectively specified probability law and representation, let X_j be the signed contribution in part j, assume finite second moments, and set $X = (X_1, \ldots, X_m)^\top$, $\mu = \mathbb{E}X$, and $\Sigma = \mathrm{Cov}(X)$. Exhaustive nonduplicated ownership gives

$$\sum_{j=1}^{m} X_j = 0 \text{ almost surely}$$

$$\iff \quad \mathbf{1}^\top \mu = 0 \quad \text{and} \quad \Sigma \mathbf{1} = 0.$$

For the forward direction, a zero total has zero mean and zero covariance with every component. Conversely, the two conditions give zero mean and zero variance for the total, so it equals zero almost surely. Each component satisfies

$$\mathrm{Var}(X_j) + \sum_{k \neq j} \mathrm{Cov}(X_j, X_k) = 0.$$

This is a joint whole-history restriction, not a requirement that every pair of episodes be negatively correlated. Near-equalities in estimated moments are not literal almost-sure closure. An empirical test must propagate joint estimation, calibration, and missing-history uncertainty and keep the complete-domain and reference-law assumptions intact. Pooling incompatible representations or selected fragments would test a different object.

Randomized remainder and delayed-treatment localization

Randomized assignment supplies a distinct complete-remainder restriction. Let Z denote assignment, L the pre-assignment signed prefix, and τ a common, prospectively fixed treatment-onset time after assignment. On the primary branch, E contains post-assignment contributions before both τ and the truth endpoint; F contains the remaining strict-pre-endpoint contributions at or after τ. If the stream ends before or at τ, F = 0. A separately declared inclusive branch adds the terminal contribution J_T :

$$R = E + F, \qquad R_{\text{incl}} = E + F + J_T.$$

Exact closure on the declared branch makes its complete remainder equal to −L. If assignment is independent of L, remainder distributions agree across arms; stratified randomization requires within-stratum comparisons or standardization to one baseline distribution. None of these statements requires a particular component or biological marker to differ. A closure-enforcing simulation must check that its selection or reweighting preserves the assignment-independence premise.

A localization prediction requires an all-intervention structural model on a common probability space. For the same pre-assignment history h and declared bearer, routing, representation, time cut, and endpoint architecture, suppose primary closure holds under both arms:

$$L(h) + E^{a}(h) + F^{a}(h) = 0,$$

$$L(h) + E^{b}(h) + F^{b}(h) = 0.$$

$$E^{a}(h) - E^{b}(h) = -\left[F^{a}(h) - F^{b}(h)\right].$$

An independently justified nonzero restriction on the entire later contribution forces an opposite early difference. Immediate treatment effects alone do not supply that restriction when subsequent experience can compensate. The inclusive branch also includes the arm-specific terminal terms in the subtraction. A directional or distributional biological prediction additionally needs a history distribution and validated observation or viability bridge; the pathwise identity alone does not identify it.

Assignment before a marker but treatment after it is post-assignment and pre-treatment, not automatically retrocausal. With valid randomization and no pre-treatment assignment channel, ordinary local forward-causal models predict no systematic assignment effect. A different prediction requires a declared global-selection, viability, whole-trajectory, or information mechanism. A marker occurring before a qualifying stream begins concerns the developmental extension rather than a measured phenomenal residual.

For a sharp formal fork, consider one candidate continuation d that ends biological life essentially immediately and another that permits ordinary continuation. Let h now denote the complete predecision history. Under the additional biological-death condition and an independent bridge placing the stream endpoint no later than death, let C_d (h) be the complete decision-to-endpoint contribution for the declared branch. A necessary compatibility condition is:

$$L(h) + C_d(h) = 0.$$

The phenomenally silent case C_d (h) = 0 requires L(h) = 0; this does not make closure sufficient for death or determine a choice probability. A separate route-selector must state how admissibility affects action and what information it uses. If route-to-contingency mapping is independently randomized and locally concealed, an ordinary local policy cannot condition on that mapping. Supplying it secretly to a simulator does not establish biological access. Section 14 develops an empirical mosquito implementation of this terminal fork, alongside its formal analysis and simulation controls.

10. Rival tournament under equal information budgets

Every candidate and rival receives the same raw variables, observation kernel, missingness inputs, data split, tuning, representation complexity, computation, and predictive-information budget (Blackwell, 1953). Prospectively chosen proper scores compare predictive laws; the multi-gate rubric remains separate. Relative rank is followed by absolute adequacy, recovery, common support, and open-set checks.

Rival models and differential pressure tests

Regulation and robust perfect adaptation can reproduce opponent, homeostatic, allostatic, or robust set-point return. The differential pressure test asks whether a held-out signed integral and terminal residual remain beyond local set-point recovery.

Computational mood can reproduce history dependence generated by recent expectations and prediction errors. The differential pressure test asks whether frozen completion potential adds predictive structure across modules.

Flexible baseline can reproduce a nonzero terminal attractor specific to subject or state. The differential pressure test compares zero-restricted and nonzero-attractor baselines under held-out proper scoring, calibration, absolute-adequacy, and common-support checks.

Conditioned bridge can reproduce paths pinned by endpoint conditioning. The differential pressure test scores event probability, the normalizer, and unexpected endpoints.

Endpoint selection can reproduce hazards that depend on latent state, frailty, care, or prognosis. The differential pressure test examines joint hazard–ledger fit and endpoint-surprise transport.

Observation floor can reproduce resolution loss, clipping, sedation, or channel shutdown. The differential pressure test uses independent channels and calibrated latent deconvolution.

Memory kernel can reconstruct retrospective balance through peak–end effects and duration neglect. The differential pressure test asks whether a prospective experience ledger retains signal after memory adjustment.

Expected neutrality can reproduce zero population or conditional means despite residual variance. The differential pressure test uses variance, atom, and pathwise analyses to distinguish means from streamwise closure.

Open-set generator can reproduce dynamics that none of the specified candidates fit. The differential pressure test examines absolute adequacy and recovery without forcing a choice.

Ordinary policy control can reproduce actions shaped by reward, effort, habit, threat, and viability. The differential pressure test asks whether a residual term improves held-out calibration beyond equal-information ordinary policy models.

Let Q be a prospectively declared rival law and $\mathcal{F}_{i,0}^{\mathrm{ex}}$ its baseline information sigma-field. Expected neutrality under Q is a weaker balance claim. With finite second moment, conditional mean zero plus positive conditional variance on a set

of positive Q -probability does not imply an atom at zero or streamwise closure.

Expected neutrality is weaker than pathwise closure

$$\mathbb{E}_Q[(Z_i^\star)^2] < \infty, \quad \mathbb{E}_Q(Z_i^\star \mid \mathcal{G}_{i,0}^{\mathrm{ex}}) = 0 \quad Q\text{-a.s.},$$
$$Q\big(\mathrm{Var}_Q(Z_i^\star \mid \mathcal{G}_{i,0}^{\mathrm{ex}}) > 0\big) > 0.$$

Model ambiguity is admissible. If two mechanisms induce the same distribution of observables under all allowed interventions, the study cannot identify their locus until an intervention or measurement separates them. Generator and verdict recovery are separate. A pipeline may rank the true generator yet issue the wrong verdict, or abstain while generator identity remains ambiguous. Sealed simulations score generator classification, verdict recovery, model ambiguity, open-set detection, and no-adequate-estimate calibration; none compensates for another.

History information versus predictive approximation

A gain from adding history can have two sources. On one held-out evaluation law, let B denote baseline information, H additional history, and Y a future outcome. Let p_H and p_0 be the true conditional laws of Y given (B,H) and B , and let q_H and q_0 be frozen predictive densities on the same outcome space. Assume the relevant log expectations are finite and the predictors give positive density on true-law support. Define

$$\mathcal{E}_H = \mathbb{E}D_{\mathrm{KL}}(p_H \| q_H), \qquad \mathcal{E}_0 = \mathbb{E}D_{\mathrm{KL}}(p_0 \| q_0).$$

Then the expected logarithmic-score gain is

$$\mathbb{E} \log \frac{q_H(Y \mid B, H)}{q_0(Y \mid B)} = I(Y; H \mid B) + \mathcal{E}_0 - \mathcal{E}_H.$$

Add and subtract the two true conditional log densities to obtain the identity. Thus history can contribute information, improve approximation, or both; predictive gain alone is not a calibrated estimate of mutual information. Shared evaluation data, frozen fitting choices, and adequately flexible rivals make the comparison meaningful. The decomposition complements proper scoring rather than replacing it (Cover & Thomas, 2006; Gneiting & Raftery, 2007).

11. Existing evidence and the scientific opportunity

Established neighboring results

Positive and negative affect can be partly separable and co-occur, supporting signed, bipolar, or vector representations (Cacioppo & Berntson, 1994; Larsen et al., 2001).

Adaptation after life events is well documented and heterogeneous (Diener et al., 2006; Luhmann et al., 2012). Opponent-process, allostatic, incentive-salience, interoceptive-prediction, homeostatic reinforcement-learning, computational-mood, and robust integral-feedback systems generate rebound, history dependence, dissociations among liking, wanting, and learning, and measurable forms of compensation (Solomon & Corbit, 1974; Koob & Le Moal, 2001; Berridge et al., 2009; Barrett & Simmons, 2015; Keramati & Gutkin, 2014; Eldar et al., 2016; Keren et al., 2021; Barkai & Leibler, 1997; Briat et al., 2016; Aoki et al., 2019). These affective findings and biological or computational control results supply distinct precedents and rival models; biomolecular adaptation is not itself evidence of a phenomenal-valence mechanism.

Experienced and remembered utility can diverge through duration neglect and peak–end weighting (Fredrickson & Kahneman, 1993; Kahneman et al., 1997). This divergence makes prospective measurement and explicit memory rivals essential. PTVC extends that foundation into a new integrative question: whether complete-stream dynamics exhibit an exact terminal boundary condition.

Plausibility signals and evidential roles

Hedonic adaptation, biological set-point control, and terminal change provide mechanism precedents, comparator models, and endpoint-study opportunities. Their PTVC-specific evidentiary weight depends on protocol-fixed, branch-matched bridges.

Constructed zero-integral functions, completion potentials, bridges, and Doob transforms establish mathematical coherence and generate testable predictions. Held-out transport determines their empirical weight.

Retrospective life evaluation supplies evidence through a memory-and-demand observation kernel and can be compared directly with prospective experience ledgers.

Endpoint proximity and changing channel variance become informative when modeled jointly with frailty, treatment, sedation, device failure, censoring, and observation floors.

How independent convergence builds confidence

The neighboring literatures contribute premises, instruments, comparator mechanisms, datasets, perturbations, and bias structures. A dependency graph makes shared samples, instruments, preprocessing, cultural ancestry, and observation channels visible so independent convergence receives appropriate weight. Confidence grows most when one frozen relation transports through distinct paths with different expected biases: prospective experience and retrospective memory; human and nonhuman tasks; survey, physiology, behavior, and

neural measurement; ordinary and terminal horizons; observational cohorts and randomized perturbations. This triangulation strategy gives more weight to methodologically independent evidence than to raw citation count (Lawlor et al., 2016; Munafò & Davey Smith, 2018).

Endpoint, horizon, and consciousness literatures

Perceived future time reorganizes motivation and emotional priorities, making disclosed horizon a strong planning rival to actual endpoint coupling (Carstensen et al., 1999; Carstensen, 2006). Time-to-death studies report heterogeneous changes in affect, symptoms, depressive burden, and psychological or social well-being near death (Vogel et al., 2013; Seow et al., 2011; Petkus et al., 2025; Wang et al., 2025). These findings support endpoint-aligned study design and provide baselines for testing declared preterminal signatures. Complete-stream measurement and terminal-zero adjudication require the additional representation and endpoint bridges developed here.

Behavioral nonresponse can conceal cognition; last report, last response, biological death, and last qualifying experience need not coincide (Casali et al., 2013; Bodien et al., 2024; Parnia et al., 2023). Comparative consciousness research cautions against limiting the eligible class to verbal humans (Birch et al., 2020). These literatures justify multimethod endpoint intervals with explicit uncertainty and provide concrete tools for testing stream individuation and eligibility across species.

Recent multicenter multimodal work in disorders of consciousness supports combining behavioral assessment, neuroimaging, and electrophysiological markers (Manasova et al., 2026). Extending this approach to PTVC requires separately validated valence and endpoint bridges; the published diagnostic and prognostic findings do not themselves identify phenomenal closure.

12. Measurement, missingness, and endpoint validity

Representation before accumulation

A confirmatory instrument must establish sign, origin, scale behavior, temporal resolution, exposure weighting, and measurement invariance across the populations and states being compared (American Educational Research Association et al., 2014; Krantz et al., 1971; Meredith, 1993). Method-diverse channels should be treated as separate fallible measurements of a latent object, not averaged until their error structure is modeled. Multitrait–multimethod validation and negative-control channels test whether common arousal, salience, motor capacity, or demand explains apparent convergence (Campbell & Fiske, 1959).

Observation coverage completes the estimand

Scheduled, randomized, participant-triggered, clinician-triggered, safety-triggered, endpoint-triggered, and device-failure observations retain separate marks. Crisis-enriched or terminally enriched samples alone do not identify a time- or exposure-weighted complete-stream integral.

Recovery requires an outcome-independent coverage component, a justified sampling-process correction with positivity, scientifically valid bounds for unsampled intervals, or an explicit partial-identification verdict.

Missing not at random is a central rival

Communication failure, fatigue, delirium, sedation, symptom severity, withdrawal, care transitions, and device loss can depend on latent valence and endpoint proximity. A 12-month burst-sampling study demonstrates that longitudinal completion and attrition patterns can be modeled explicitly and supplies a useful design template for the present missingness analysis (Prochnow et al., 2025). Selection, pattern-mixture, joint longitudinal-event, inverse-probability, and bounding approaches should be compared across prospectively specified sensitivity ranges rather than relying on complete-case or last-observation-carried-forward defaults (Rubin, 1976; Scharfstein et al., 1999; Little & Rubin, 2019).

No-lookahead and reactivity

Endpoint classifiers, smoothed latent states, reserve proxies, and learned representations may not use future observation availability or terminal labels unless the analysis is explicitly retrospective. The burden and emotional effect of measurement enter the trajectory as intercurrent events. Every cumulative proxy must pass a no-neutrality-by-construction simulation.

Positive observation probability supports recovery only with an identified observation process and measurement bridge. Valid finite bounds for unobserved exposure generally provide partial identification; point identification requires the resulting target set to be a singleton.

Replicate channels and recoverability

Method-diverse channels can identify or bound a latent distribution only under explicit independence, loading, location, moment, and support assumptions. Two-replicate identification in the spirit of Kotlarski (1967) is available only under its own additive-error and independence conditions; analyst separation and instrument diversity do not prove stochastic independence. Every bridge is first stress-tested in sealed simulations containing assumption violations, shared causes, response shifts, shutdown, and state-dependent reporting. A favorable observed result first constitutes measured-scale evidence and advances to latent PTVC evidence as the protocol-fixed latent gates are satisfied. The Day Reconstruction Method improves structured within-day coverage but remains a retrospective reconstruction whose recall and selection process belongs in the observation model (Stone et al., 2006).

Randomized ecological momentary assessment (EMA) studies show that prompt schedule and response format can alter measured affect distributions or dynamics. Prompt cadence, response format, anchors, and interface defaults therefore belong inside the observation kernel and require frozen or calibrated transport rules (Langford et al., 2026; Haslbeck et al., 2025).

Shared measurement error can itself mimic correction. In discrete time, suppose zero-mean errors e_k are mutually independent, independent of the latent process, and have common variance σ^2. For fixed weights w_k and a positive integer lag h,

$$\operatorname{Cov}\left(\sum_{k\le t} w_k e_k,\; e_{t+h} - e_t\right) = -w_t\sigma^2.$$

Only the common current error contributes: it enters the estimated prefix positively and the later observed change negatively. A negative prefix-change association can therefore arise even without compensatory latent dynamics. A calibration model or independent replicate channel should account for this term before interpreting the association as evidence for a route. Serially dependent or heteroskedastic errors require their actual covariance terms instead of this special-case simplification.

13. Confirmatory decision structure and model recovery

Fix deterministic verdict rules before outcome access. The reporting vocabulary below constitutes the required confirmatory record: it separates the four-axis public result vector from supporting prerequisite, specification, recovery, and attainment fields, and its conjunctive gates prevent one passed module from masking another’s failure.

Confirmatory reporting and decision vocabulary

Prerequisites are recorded as valid, failed, or unavailable. The prerequisite checks cover representation and sign; neutral origin; the protocol-fixed scalar or non-scalar branch; exposure rule and clock; qualifying bearer; ownership and routing; complete strict-pre-endpoint domain; truth-level endpoint; observation and missingness; and the measured-to-latent bridge. A failed or unavailable prerequisite yields No adequate estimate / invalid gate and blocks downstream promotion.

Target specification identifies the geometry as exact residual, finite band, or atom at zero. The declared modality is [SET-ALL], [P1] (probability-one), [POLICY-ALL], [SUPPORT], [CONTINUATION-ALL], or [FINITE-CLASS]. TargetStatus uses the controlled verdicts Equivalent, Different / off-band, Informative but inconclusive, No adequate estimate / invalid gate, or Target-adverse, matched to the specified target. Exact-atom evidence remains nested and branch-matched.

Recovery status records generator recovery, scientific-verdict recovery, and absolute adequacy. Recovery or availability

failure yields No adequate estimate / invalid gate. Failure of the declared open-set check yields Open-set inadequate.

RouteStatus and ComparatorStatus use the controlled verdicts Route-favored, Rival-favored / Route-failed, Model-ambiguous, Open-set inadequate, or No adequate estimate / invalid gate. Target-adverse remains a TargetStatus result. Equal-budget scoring, calibration, absolute adequacy, common support, and open-set checks govern promotion.

Attainment status distinguishes fixed-policy probability-one completion, almost-sure attainability, limit-sure only, positive-probability only, and unavailable attainment. These route-policy quantifiers do not replace the target-specification modality.

Inference status identifies whether the result is identified-set, frequentist, Bayesian, or sensitivity-based. The record preserves its evidence-object type, assumptions, coverage or prior basis, and sensitivity guarantees. Controlled public verdict labels remain on their corresponding axes without cross-promotion.

Recovery before interpretation

Synthetic and semi-synthetic studies span PTVC routes, expected-neutrality laws, diffuse and atomic residuals, flexible baselines, conditioned bridges, state-dependent hazards, regulatory and computational models, memory reconstruction, selection, censoring, heavy tails, stochastic volatility, and observation floors.

A correctly computed empty inversion or feasibility set is not NAE and receives its protocol-fixed object-matched state: model-incompatible, bridge-failed, route-failed, or another named rejection. NAE is reserved for unavailable, undefined, invalid, or uncomputed objects. A protocol-fixed recovery matrix reports generator classification and scientific verdict by generator class, including worst-class performance. The pipeline controls false favorable support and false counterexample declarations.

Negative controls and open-set alternatives

Negative controls test leakage and common-cause explanations (Lipsitch et al., 2010). If every offered model fails absolute adequacy, report Open-set inadequate; adequate but inseparable alternatives yield Model-ambiguous. Unavailable or invalidly recoverable target information yields No adequate estimate / invalid gate. Open-set generators prevent forced selection among misspecified candidates. Proper scores and calibration accompany scientific margins rather than override these distinctions (Gneiting & Raftery, 2007).

Protocol versioning preserves every result: later refinements remain traceable to the version under which each gate and counterexample was evaluated.

Unresolved cases remain visible

Unresolved bearer, endpoint, observation, or bridge cases cannot disappear from the denominator. Use one fixed target, modality,

eligible frame, and weighting rule. The displayed interval is a sharp deterministic bookkeeping bound only when favorable and adverse classifications are truth-containing under the same gate system. Confidence-set versions require a simultaneous coverage statement; posterior and sensitivity versions retain their protocol-specified interpretations. Without such a guarantee, report classification bookkeeping rather than a bound on the true violation prevalence.

Transparent unresolved-case bound

$$W_E := W_F + W_A + W_U > 0,$$
$$\frac{W_A}{W_E} \leq p_{\text{viol}} \leq \frac{W_A + W_U}{W_E}.$$

Let W_F be weighted favorable classified mass, W_A weighted adverse classified mass, W_U weighted unresolved or gate-invalid mass, W_E total eligible mass, and p_{viol} the eligible-population violation fraction. This interval is deterministic bookkeeping, not a confidence or posterior interval unless the protocol-fixed simultaneous inferential guarantee supplies that interpretation.

14. Experimental program: calibration and mosquito terminal-choice tests

The experimental program combines representation and local-route studies with a contained mosquito terminal-choice study. The latter tests a specified action-selection extension of PTVC and a separate future-regime developmental prediction. Calibration, apparatus validation, and model-recovery work support the empirical experiments; terminal experiments are not restricted to simulations. Each study begins with the assumptions, measurements, and independent review appropriate to its own question.

Stage 0 — protocol fixation, public preregistration, and rigorous design

Freeze the target, measurement, route, rival, scoring, error-control, abstention, retirement, and software choices listed in the protocol. Conceal generator labels from analysts and adjudicators until the simulation pipeline is locked. Where roles permit, conceal empirical outcomes until the empirical pipeline is locked. Document the two procedures separately.

Stage 1 — representation and local complementarity

Use safe bounded episodes with randomized benign assignments across valence modalities, controllability, predictability, duration, and horizon, drawing on micro-randomized trial design (Klasnja et al., 2015). Under consistency, well-defined versions, positivity, and the declared interference structure, randomization identifies assignment effects, not delivered, experienced, or latent phenomenal dose. Those dose variables remain separately

calibrated with explicit noncompliance and measurement bridges. Test whether an independently measured future response area complements accrued area at randomized cuts against equal-budget ordinary rivals. A transported local cancellation result would support episode-scale route calibration and prepare the next prospectively specified terminal test.

Stage 2 — frozen completion-geometry transport

Estimate one completion potential on training modules and freeze it. On held-out modules, test transition calibration, conditional increments, action odds under randomized information, future-area prediction, residual compression sufficiency, and resolution transport. Off-support actions return partial identification or No adequate estimate. Model selection and representation learning remain inside the training loop.

Promotion criterion

Promotion requires generator and verdict recovery, stable origin and measurement, held-out calibration, superiority to equal-budget rivals, correct negative-control behavior, and replication across a site or channel. Proper scores evaluate predictive distributions; the conjunctive vector remains a decision rubric. Passing supports only the local route and progression to terminal studies.

Immediate analyses in existing data

Before new endpoint studies, existing longitudinal resources can immediately test foundational pipeline components. A measured ledger proxy must be built without future endpoint information. At each prospectively specified landmark, all representations, covariates, competing-risk forecasts, and tuning are fit out of fold using data available by that landmark; realized time and cause are then scored before route-by-surprise tests. Held-out compensator-residual and time-rescaling diagnostics assess calibration under a construction matched to the protocol-specified ordinary, marked, or cause-specific, terminating, and right-censored endpoint process (Cox, 1972; Andersen et al., 1993; Brown et al., 2002). Cause-specific intensity, cumulative incidence or event probability, and interventional endpoint risk are distinct estimands. Thus the test is whether one frozen relation outpredicts selection, local control, prognosis, and hidden-state rivals, not whether hazard merely declines near zero.

Terminal-dispersion analyses must decompose observed variance into latent state, observation error, and their covariance; narrowing can reflect true contraction, censoring, scale compression, sedation, or channel loss. Cohorts with dense affect and later global evaluation can separate experienced value from peak-end, duration-neglect, and narrative-memory rivals. Cross-dataset analyses should retain dataset-specific measurement models and use leave-one-dataset or leave-one-country-out transport rather than force a common

coefficient. Causal transport requires explicit invariance assumptions rather than coefficient similarity alone (Pearl & Bareinboim, 2014).

Analyses comparing anticipated with unexpected endpoints freeze a competing-risk forecast at a prospectively specified landmark, score realized time and cause, and only then test route-by-surprise interactions.

Three high-value experiments

First, prospectively specified nonanticipating stopping kernels randomized from frozen history probe occupation functionals on a protocol-fixed basis. Full recovery of unrestricted basis coefficients requires separation; particular functionals or informative bounds can remain available without it. Require well-defined intervention versions, consistency, no interference or a declared interference map, known assignment or conditional exchangeability of future potential outcomes, positivity on common support, and declared observation and censoring models. Report basis, rank, condition number, and null space; distinguish physical stopping from disclosed horizon. Ill-conditioned recovery requires stability checks; report identified functionals or valid bounds, and use NAE when neither is available for the declared target. Second, random-cut complementarity with benign appetitive and aversive micro-interventions separates bookkeeping from causal compensation while assigned, delivered, experienced, and phenomenal dose remain distinct. Third, freeze one completion potential across

transition, action, endpoint, and terminal-support modules. All three combine empirical testing with sealed simulation benchmarks, open-set generators, and branch-matched interpretation.

Mosquito terminal-choice and future-regime experiment

The central proposal is an empirical test of whether an organism's prior conditions and the time available after a salient experience predict its choice of a continuing-life route at a terminal fork. Begin with 300 mosquito eggs, randomly assigned before hatching to three future adult-history groups of 100. The groups later receive normal, enriched, or adverse conditions. Individual mature adults then encounter twenty equivalent outlet entrances: one leads to ordinary continuing conditions and nineteen lead to a terminal apparatus. The pre-hatching assignment also permits a separate test of hatching and developmental outcomes before the assigned adult histories begin. This is a proposed study, not a report of observed results.

Species, developmental stages, and realistic timing

Use a specified noninfected laboratory strain of *Aedes aegypti* in a contained insectary. Eggs hatch into aquatic larvae, not flying adults; larvae pass through the pupal stage before adult emergence. The CDC describes approximately seven to ten days from egg to adult under favorable conditions, while actual timing depends on the colony and its conditions (Centers for Disease Control and Prevention, 2024). Distinguish egg

hatching, adult emergence, chronological age, adult age, and treatment duration in the records. Egg-storage age is not a substitute for developmental age. The experiment therefore cannot place newly hatched mosquitoes in a flight-choice chamber six hours after hatching.

Following adult emergence, maintain a common seventy-two-hour maturation period before any assigned adult-history exposure. This is a proposed design choice informed by evidence that flight performance matures over the first one to three adult days, not a claim that all mosquitoes share an exact developmental deadline (Briegel et al., 2001). Verify ordinary flight and handling performance in independent calibration cohorts. Published laboratory methods provide the species-specific husbandry and containment foundation (Wohl & McMeniman, 2023). Prespecify sex and mating status, use balanced allocation across emergence batches, and analyze sex-specific responses rather than assuming identical physiology. The study uses sugar-resource manipulations and does not require human feeding, pathogen infection, or environmental release.

A biologically realistic duration comparison tests adults at six days after emergence. Assign adult-history durations of 24, 48, or 72 hours. For the 24-hour condition, begin the history on adult day five; for 48 hours, on day four; for 72 hours, on day three. Keep the preceding period under the same common husbandry. All groups then reach the chamber at the same adult age and prespecified circadian phase. This separates duration from the simple difference between a younger and an older

mosquito. Retain the actual elapsed times and pre-treatment histories, since changing onset time changes an exposure history, not an abstract dose alone.

The initial 300-egg cohort should evaluate one prespecified duration and post-event schedule, for example the 48-hour adult-history condition. Further independently randomized cohorts evaluate the other durations; the original three groups should not be subdivided into a large factorial grid with only a few mosquitoes per cell. Under the illustrative seven-to-ten-day developmental interval, a six-day-old adult is tested roughly thirteen to sixteen days after development begins. The actual emergence record governs eligibility. A fixed calendar date must not force late-developing or immature animals into a flight assay.

Allocation before hatching and the three histories

Record the 100/100/100 future assignments before hatching using a concealed randomized allocation. Block across parental or egg-laying batches where feasible. All eggs, larvae, pupae, and newly emerged adults receive equivalent physical conditions until their assigned adult treatment begins. Balance containers, density, water, food, handling, position, temperature, humidity, and light exposure, and record departures. Use coded assignments so staff maintaining the common developmental environment need not know future history labels. Multiple independent containers per history prevent a treatment effect

from being indistinguishable from one cage or one position effect.

Group 1 receives normal laboratory living conditions. Group 2 receives enriched conditions selected to provide favorable temperature, reliable food, and other species-relevant opportunities. Group 3 receives an adverse-history condition involving less favorable, prospectively specified resource or environmental circumstances. The operational differences and exposure limits are fixed through species-specific calibration and review before confirmatory outcomes are examined. Do not combine arbitrary extremes of temperature, deprivation, and handling merely to intensify a label; define which contrast is being tested and measure flight capacity and pre-choice attrition separately. The words normal, enriched, and adverse name assigned environments. Whether their net phenomenal contributions have the intended sign is a measurement question, not something established by the labels.

The design begins with 100 eggs per history, not a guarantee of 100 adult choices. Retain counts for eggs allocated, eggs hatched, larvae reaching pupation, adults emerging, adults reaching the scheduled test, and adults producing an observed choice. Record pre-choice mortality, nonchoice, handling failures, and observation losses without substituting animals after outcomes become known. Container-level treatment makes the container an experimental replicate; mosquitoes sharing it are nested observations (Lazic et al., 2018).

The twenty-outlet chamber and the five-percent null

Test one mature adult at a time in an enclosed chamber with twenty equivalent entrance apertures. One outlet leads to normal continuing-life conditions; the other nineteen lead to a contained terminal apparatus. The proposed implementation retains the mosquito-zapper concept, subject to independent validation of the actual terminal procedure, its latency, and its suitability for the proposed study. The relevant decision is the first prospectively defined committed outlet entry before feedback or contact with the downstream apparatus. Record route entry, subsequent survival, verified terminal events, and their times as separate outcomes. Entry into a nominally terminal outlet is not itself proof of an immediate conscious-stream endpoint.

Choose the continuing-life outlet uniformly and independently for each trial and conceal its physical identity. This balances positions in expectation. A separately chosen blocked allocation must instead record and analyze its actual conditional assignment probabilities. Match geometry, illumination, airflow, humidity, odors, sound, heat, vibration, and any apparatus-associated electrical or chemical cues at the choice point. A visually identical entrance is not enough. Use nonterminal apparatus trials to quantify residual side preferences and cue leakage, with blinded scoring and a frozen route-to-contingency record. The construction must prevent downstream events from broadcasting informative cues to later trial animals. These checks validate a biological choice assay; they do not replace its terminal condition with a simulation.

Let $D=1$ indicate a recorded committed choice before the fixed deadline, let C be that outlet, and let K be the randomly assigned continuing-life outlet. Under the null, require K to be uniform and independent of both the choice and its recording status, conditional on the declared pretrial design information. For a nonzero choice probability, the null therefore gives

$$\Pr(C=\mathrm{K}\mid D=1)=\tfrac{1}{20},\qquad \Pr(D=1,C=\mathrm{K})=\frac{\Pr(D=1)}{20}.$$

The first equality follows by summing over the twenty possible recorded choices; the second follows by multiplication by the choice probability. Stable physical-side preferences do not change this result when the concealed mapping remains independent. Among all tested animals, however, the baseline for a recorded safe choice is five percent times the probability of recording a choice, not automatically five percent. Nonchoices and observation losses remain separate outcomes. Apparatus checks quantify detectable cue leakage and side effects; they cannot establish independence merely from visually identical apertures.

Use matched all-nonterminal sham trials in which every outlet leads to ordinary continuing conditions and one is independently labeled as the target for analysis. These preserve the twenty-outlet measurement problem while removing the terminal contrast. The hypothesis of interest is a history-related enrichment that depends on the real downstream contingency, not merely on altered locomotion, general escape behavior, or

increased willingness to choose an outlet. Such sham trials are experimental controls, not a replacement for the proposed terminal arm.

Temporally decoupled post-commitment arm. A second, prospectively separated design makes every entrance lead into a matched passive holding compartment. A video-verified first committed entry and compartment closure precede a freshly generated uniform label K identifying the single continuing-life assignment. The control process logs the commitment timestamp, closure timestamp, randomization input, generated K and subsequent outcome. The intended separation removes any K-dependent pre-choice apparatus difference: the designation does not yet exist when the animal commits. It does not, however, establish that the later designation was statistically independent merely because the hardware looks identical. The randomness, recording logic, segregation of operators and outcome handling must be independently audited.

The timing distinction changes the research question. In the original pre-choice-mapping arm, K exists before commitment and a route-selection mechanism is evaluated while cue and mapping leakage remain empirical rivals. In the post-commitment arm, no entrance has a pre-existing safe identity at choice time; a later random outcome is attached to the already committed compartment. Any PTVC-related prediction of a C–K association in that arm consequently requires a separately declared whole-trajectory, nonlocal, or global-selection extension. It does not follow from exact terminal valence closure or from the pre-choice selector alone. A discrepancy would first

challenge the joint independence model and demand independent investigation of randomization, timing, data integrity and dependence, rather than uniquely identify PTVC.

Sharp random-label null. Write C for the committed compartment, D=1 for a valid recorded commitment, and H for any pre-randomization physiological or exposure history. Under genuinely uniform post-commitment randomization with K independent of (C,D,H), the conditional match probability is 1/N, however uneven the distribution of C or the probability of D=1. This is a probability identity under an independence model, not a proof that observed randomization implements that model. A deviation cannot be attributed solely to hunger, fatigue, side preference or survivor composition if the conditional independence has actually been established. The all-tested recorded-match probability is P(D=1)/N; nonchoice and attrition remain separate outcomes.

$$K \sim \text{Unif}\{1, \dots, N\}, \quad K \perp (C, D, H) \implies \Pr(C = K \mid D = 1, H) = \frac{1}{N}.$$

Outlet-number transport. The twenty-outlet design remains the first behavioral assay (null 5%). A one-hundred-outlet replication (null 1%) is a separately calibrated extension after independent verification that adults can actually discriminate and commit among that many entrances. A ten-thousand-outlet arrangement has a formal 0.01% null but is an engineering thought experiment until equivalent accessibility, containment, masking, valid-choice recording and realistic sample size are demonstrated. Increasing N also changes geometry, search time, nonchoice, route availability and the information available to the

animal; the claim under test must transport across those changes rather than treating N as a cost-free denominator multiplier.

Experimental separation and confirmation. Analyze the pre-choice and post-commitment arms as distinct preregistered tests, with different randomization timestamps, validity gates and likelihoods. Repeat masked all-nonterminal sham and randomized-label controls; compare history-by-duration and history-by-feeding-delay interactions within each arm. The initial 300-egg allocation is a feasibility cohort, not a justified sample size for a definitive 20-laboratory test. Model original egg assignment, independently randomized housing units, adult-emergence attrition, recorded commitment and laboratory variation as separate levels. A downstream analysis that discards failed choosers or aggregates animals from one cage as independent trials changes the estimand.

What the Law-specific prediction requires

The Law-specific composite model makes a concrete, testable prediction. Suppose the animal carries a qualifying stream; the assigned history changes the distribution of unresolved signed contributions; the declared biological-death condition applies; the terminal route cannot supply the required remaining contribution before the relevant endpoint; and an action-selection route can restrict choices accordingly. Under this composite model, animals whose histories remain incompatible with the terminal routes should be more likely to select the single route permitting continued life. Enriched and adverse

histories can both produce such a prediction if each creates a larger fraction of unresolved streams than normal conditions. Their direction and magnitude are fixed before data inspection.

The mosquito need not know a mathematical ledger or consciously recognize a route that permits continued life. A proposed whole-trajectory mechanism could influence which impulses and actions are realized while behavior remains spontaneous. A concealed mapping makes the test unusually informative: the model must specify how its predicted choices relate to a continuation contingency unavailable to ordinary local sensory policies. Pre-choice concealment, the separate post-commitment design, and matched sham trials place different pressures on that account. Sensory, physiological and global-selection alternatives can be compared under the same frozen information and scoring rules.

A useful illustrative mixture makes the numerical prediction explicit. Among animals with a recorded choice, let u be the proportion for which the terminal routes are incompatible under the declared model, let s be their probability of selecting the continuing-life route, and let q0 be the calibrated probability among the remaining animals. Then the group probability is the mixture below. The strongest one-safe-route version sets s = 1. With q0 = 0.05, rates of 15–25% correspond to u of about 10.5–21.1% when s = 1. For a confirmatory prediction, u, s and q0 must be independently calibrated or bounded before matching outcomes are examined, then held fixed across the declared duration and outlet-count comparisons. The illustrative rates are study-design alternatives rather than measurements of mosquito

experience. If all recorded choosers meet the strongest incompatibility assumption, it predicts a 100% continuing-life choice rate.

$$p_{\text{safe}} = (1 - u)q_0 + us;$$
$$s = 1,\ q_0 = 0.05 \Longrightarrow p_{\text{safe}} = 0.05 + 0.95u.$$

The terminal contribution cannot be set to zero merely because the apparatus acts quickly. Section 9's terminal-compatibility equation retains any decision-to-endpoint contribution on the declared domain. A finite bound on that contribution can make a terminal route incompatible with a larger independently bounded prefix; an unmeasured contribution leaves the branch less identified. Equally, closure does not require an organism to die once balance is reached. Continued life after an intermediate zero is compatible with the boundary hypothesis. These distinctions connect the experiment to the existing formal target without changing it to fit a desired outcome.

Timing after feeding or another salient event

Adult-history duration and time after a salient event are distinct variables. In an additional series, hold adult age, history duration, and chamber circadian phase fixed while randomizing standardized sugar-feeding opportunity schedules. A proposed starting set places the end of that opportunity 15 minutes, 2 hours, 6 hours, or 24 hours before chamber entry. These are nominal assigned lags for mature adults, not egg-to-flight intervals or established times of maximal pleasure. Record actual ingestion, its completion time, refusal, consumed amount,

and subsequent resource access. The primary comparison follows the assigned schedule. Analyses indexed by actual ingestion time or dose must address selection into feeding. An event-triggered alternative can randomize a delay after observed ingestion, but it then needs its own age and circadian design rather than claiming those clocks remain automatically identical.

The timing conjecture is that a recent positive experience could temporarily increase an unresolved positive contribution, leaving less time for an opposite contribution before the terminal fork. Under a specified transient-compensation model, an earlier chamber encounter would therefore produce greater safe-route enrichment than a later encounter. This is the experimentally meaningful version of the proposed “peak pleasure at 1 hour 32 minutes, test at 1 hour 35 minutes” example: identify the event independently and vary the interval after it, rather than assume a particular clock time reveals a subjective peak. The same logic can be applied to another independently calibrated appetitive event, with its own preregistered timing and measurement model.

For the positive-residual branch, let t_0 be the event cut and let Δ bound the entire interval from that cut to the relevant stream endpoint on a terminal route. It must cover the waiting period, chamber choice time, and apparatus-to-endpoint latency, not merely time to chamber admission. On an absolutely continuous contribution branch with opposite-sign rate bounded by M, the total negative capacity is at most $M\Delta$. Any allowed atomic or endpoint-inclusive contribution needs an additional bound in that capacity. Under these premises, the

following directional obstruction applies. The bound is a conditional design requirement, not an established measurement of pleasure or a known mosquito compensation rate. Without independently calibrated capacity and endpoint bounds, the delay comparison tests the specified temporal pattern but does not establish insufficient capacity from elapsed time alone.

$$L(t_0) > M\Delta \Longrightarrow L(t_0) - \mathrm{Cap}^{-}(t_0, t_0 + \Delta) > 0,$$
$$\mathrm{Cap}^{-}(t_0, t_0 + \Delta) \leq M\Delta.$$

Timing becomes a discriminating prediction when an independently calibrated transient-compensation model specifies which history and event should yield higher early safe-route enrichment. A positive event might increase an existing positive residual or reduce an earlier negative one; the proposed direction therefore follows the declared path model. Freeze the early-versus-late contrast before data collection and test the complete history-by-delay relationship. This makes elapsed time a genuine experiment on the candidate mechanism, rather than a retrospective interpretation of the interval with the largest observed effect.

Feeding also changes ordinary physiology and behavior. Blood feeding can change host attraction, while olfactory sensitivity and activity vary over the daily cycle (Dou et al., 2024; Eilerts et al., 2018). Thus satiation, flight energy, meal mass, motivation, digestion, and circadian phase are explicit alternative explanations. Keep age and chamber phase matched where possible, rotate schedules across independent blocks, and measure locomotor competence in separate control cohorts.

Varying a feeding delay can change the phase of feeding even when the phase of choice is fixed; record both. A recently fed mosquito that does not move is not evidence that it selected the continuing-life route.

Hatching and adult-emergence predictions

Because future history assignments are recorded before hatching while developmental conditions remain equivalent, the experiment also tests whether early developmental outcomes differ by those assignments. The primary early endpoint can be the proportion of the original eggs hatching within a fixed observation window. Record time to hatching, later adult emergence, and developmental attrition separately. A hatching difference is not interchangeable with an emergence difference. The window and treatment-onset schedule must ensure that the future adult history has not begun when the early endpoint is measured.

Under an ordinary forward-causal null with valid randomization and no physical pathway from the future assignment to incubation, hatching probabilities should not systematically depend on that assignment. The proposed developmental extension asks whether a whole-trajectory or viability constraint changes which individuals enter particular future circumstances. For example, it may predict fewer successful hatchings in the enriched or adverse future groups when those trajectories would otherwise encounter the terminal fork with unresolved contributions. This is a separate,

prospectively specified extension; the primary PTVC equation does not by itself calculate a hatch-rate difference or establish consciousness in an egg.

A credible early effect must survive checks for parental and batch imbalance, container microclimate, label leakage, handler behavior, egg age, fertilization, and differential observation. Freeze one primary developmental contrast and its direction. Use randomization-based comparisons respecting the actual allocation blocks, and repeat with new independently handled batches and analysts. Post-outcome placebo labels can diagnose analysis errors but cannot rule out a physical cue or allocation channel present in the original experiment. Neither an early difference nor a later choice difference, on its own, proves influence backward in time.

Analyze developmental outcomes by original assignment, including eggs that never reach the adult experiment. A model indexed by the future regime actually delivered must separately address cancelled schedules, crossover, non-delivery, and failure to reach adulthood (Hernán & Robins, 2020). Conditioning only on successful emergence can select different animals in different groups. Report safe-route choices both relative to the number of tested adults and as part of the complete assigned-egg outcome sequence. Do not count hatching and choice results from the same selected cohort as independent confirmations.

Analysis, repetition, and decisive interpretation

The primary adult comparison must name its denominator. Prespecify history contrasts in the probability of a continuing-life choice among recorded choosers, and report the separate assignment-based probabilities of a safe choice, a terminal-route choice, and nonchoice among all tested adults. A chooser-conditioned history contrast can reflect selection into choosing; it is not automatically the intention-to-treat effect of history. The calibrated five-percent null applies to the original recorded-chooser estimand under its mapping assumptions; the post-commitment arm uses the corresponding one-in-N null and its own later-randomization independence test. Report absolute rates, risk differences, interval estimates, choice latency, and verified post-choice outcomes. Control the planned family of history, duration, delay, sham-interaction, and developmental comparisons. Section 16 supplies a sequential option for repeated cohorts; exploratory patterns require a new confirmatory cohort.

For 100 independent recorded choices in a twenty-outlet test, the exact 5% null expects five matches. A one-sided exact binomial test at a nominal 5% level rejects that null at ten or more matches (actual null-tail probability about 0.0282); its power is approximately 94.5% if the true match probability is 15%. This striking contrast is a calculation for independent recorded choices, not a power guarantee for the proposed egg cohort. Hatching, adult emergence, nonchoice, shared cages and batch effects reduce or alter the experimental information. Set

the final sample size from the true assignment unit, recorded-choice rate, overdispersion, intended contrasts and multiplicity. Three cages with 100 mosquitoes each do not supply 100 independent treatment replicates per group. Randomization, masking, exclusions and experimental-unit reporting follow ARRIVE (Percie du Sert et al., 2020).

An exceptionally informative result would combine replicated above-null safe-route matching with the prospectively predicted differences across adult history, exposure duration and post-feeding delay. It would survive blinded apparatus checks, independently audited randomization, matched sham trials and strong behavioral comparators. Reproducing that complete signature across independently constructed laboratories would provide powerful evidence for the frozen composite route. Its implications for the primary PTVC equality would depend on the separately justified phenomenal-history and endpoint bridges. A null or reversed predeclared pattern would instead constrain that mechanism at the achieved precision. The criteria are fixed before the trial, not chosen after seeing which histories succeed.

The experiment therefore addresses two connected questions: whether terminal-contingent outcomes track a proposed lifetime-balance constraint, and whether action selection reflects processes outside reflective awareness. Its directly observed endpoint is mosquito behavior and development. The further philosophical question is whether apparently spontaneous impulses could be constrained before conscious deliberation; mosquito choices alone cannot decide the origin of human

thoughts or settle free will. By keeping the behavioral prediction precise, the study gives that broader question a concrete empirical starting point.

Implementation requires a contained insectary, noninfected laboratory mosquitoes, a reviewed environmental-history protocol, a validated terminal procedure, and documented endpoint classification. No human exposure, pathogen manipulation, or environmental release is part of the proposal. The all-nonterminal calibration and sham arms establish apparatus behavior; the terminal arm remains an empirical part of the proposed study. Archive the randomization, outlet mappings, timestamps, raw recordings, exclusions, and analysis code so an independent team can reproduce each stated contrast.

15. Endpoint-aligned extension and complete-stream adjudication

Observational bridge studies establish the path to terminal tests

Existing longitudinal cohorts and low-burden clinical data can evaluate measurement invariance, time-to-endpoint alignment, missingness models, endpoint revision, residual variance, and rival recovery. These studies establish feasibility, informative bounds, and recoverability, and can test whether declared signatures remain distinguishable from symptoms, function, prognosis, and channel shutdown. Identification of phenomenal closure additionally requires the complete-domain and latent-measurement bridges.

Prospective endpoint architecture

A terminal protocol needs serial multimethod assessment, an independent endpoint committee, immutable status history, competing-event and return logic, device and communication-quality records, medication and care context, and a joint endpoint–observation–missingness model. The endpoint predicate is fixed without ledger outcomes. Observation burden is minimized and never allowed to interfere with treatment or comfort.

Primary estimands

The primary estimands include a latent terminal-residual compatibility set accounting jointly for origin, endpoint,

prefix, suffix, observation kernel, measured-to-latent bridge, and numerical uncertainty.

A second estimand concerns terminal residual atom location and mass compared with no-atom-at-zero alternatives. The alternative class includes off-zero atoms, lattice or quantized structure, singular-continuous mass, absolutely continuous laws, and mixtures.

Endpoint-dependence and hazard-weighted occupation diagnostics are additional estimands, including stress tests of unexpected endpoints.

Finally, the program estimates transport of the frozen completion potential and local signed-dose calibration into endpoint-aligned modules.

Evidence requirements for complete-stream adjudication

Complete-stream adjudication applies the joint prefix–observed–suffix rule from Section 4: the truth-containing object, plus an adjudicated endpoint or validated terminal-remainder bridge, must locate the full latent residual against the protocol-fixed exact or finite-band target. Until then, separate bounds, protocol endpoints, and measured-scale results remain staged inputs.

Human clinical studies preserve care and comfort. No human intervention may induce harm, delay care, or manipulate dying to test PTVC, and the hypothesis is not clinical guidance. The contained mosquito experiment described in Section 14 is a distinct animal-research branch with its own prospectively reviewed terminal procedure and endpoint validation.

Endpoint surprise and promotion

A prospective program distinguishes disclosed horizon, predicted endpoint hazard, and realized time and cause. One route may work only when participants can plan; another may predict structure under concealed or unexpectedly rapid endpoints. Promotion requires the same measurement, recovery, comparator, common-support, and truth-bridge gates used to declare adversity; favorable and adverse claims receive symmetric scrutiny.

Quantitative evidence for a fully specified choice model. For independent recorded commitments with valid independent label allocation, a fixed-null match probability $p_0=1/N$, a prespecified alternative p_1, n recorded choices and k matches, the point-model likelihood ratio is the binomial likelihood under p_1 divided by that under p_0. In decimal log form it is the display below. It is also a Bayes factor only for two fully specified point hypotheses with their within-model probabilities fixed in advance. It is not the Bayes factor for the whole PTVC framework against all biological and methodological alternatives.

$$\log_{10}\Lambda = k\log_{10}\left(\frac{p_1}{p_0}\right) + (n-k)\log_{10}\left(\frac{1-p_1}{1-p_0}\right),$$

$$p_0 = \frac{1}{N}.$$

For illustration only, suppose twenty genuinely independent laboratories each produced one thousand independent valid-choice observations with N=100, and all together recorded

k=5,000 matches among n=20,000 choices. With p_0=0.01 and the prespecified point alternative p_1=0.25, $\log_{10}\Lambda$ is approximately 5,181, not 622. Under a hypothetical 99% matching rate it would be approximately 39,114, not 3,949. These enormous values compare those two point models only. The observations are not reported results or effect-size forecasts. Shared apparatus, cage or batch dependence; varying choice rates; selection into recording; data-dependent alternatives; multiple laboratories; and post-hoc outcome searches invalidate the independent binomial simplification unless the model is redesigned to account for them.

The pre-choice design tests whether mosquitoes select a secretly designated continuation under verified physical concealment. The post-commitment design tests whether committed choices are statistically associated with a subsequently randomized outcome. A confirmed departure from the latter null would be extraordinary evidence against the stipulated joint independence, but it would not alone prove which dependence mechanism generated it. RouteStatus can change only for a prospectively specified route whose auxiliary premises and recovery checks survive; TargetStatus for the universal terminal-closure statement requires its separately justified phenomenal, bearer and endpoint bridges.

16. Exactness, atoms, and symmetric error control

Exact zero is a precise boundary property. Its evaluation separates location, spread, atom mass, and the observation process while accounting for finite resolution, measurement error, smoothing, selection, and model flexibility.

Atom at zero versus no atom at zero statistical branch

$$P_i^Z(dz) = \pi_{0,i}\delta_0(dz) + (1-\pi_{0,i})G_i(dz), \quad G_i(\{0\}) = 0, \quad 0 \le \pi_{0,i} \le 1.$$

For stream i, the no-atom-at-zero alternative G_i satisfies $G_i(\{0\})=0$ and may contain off-zero atoms, lattice or quantized structure, singular-continuous or absolutely continuous mass, and mixtures. The finite-band branch asks whether $P_i^Z([-\varepsilon, \varepsilon])$ meets a protocol-fixed threshold; the atom branch asks $\pi_{0,i}>0$; probability-one closure asks $\pi_{0,i}=1$. A location-free rival estimates stream-specific atom location a_i and mass $\pi_{a,i}$; its remainder $G_{i,a}$ assigns zero mass to $\{a_i\}$. Atom existence and zero location are therefore scored separately. [SET-ALL] is the stronger deterministic universal claim.

Boundary-aware inference

Both $\pi_{0,i}=0$ and $\pi_{0,i}=1$ are nonregular boundaries, so ordinary interior asymptotics may fail (Self & Liang, 1987); at $\pi_{0,i}=1$ the remainder law G_i is unidentified. Recovery should include parametric bootstrap or valid branch-specific methods, atomless laws with high density near zero, near-unit mixtures, location-free atoms, singular alternatives, rounding, heteroskedastic error, missingness, and finite-sample stress tests.

Nested-resolution discipline

Prospectively nested bands may increase discrimination only while the latent target, acceptance sets, bridge family, recovery guarantees, and limiting rule remain fixed. Passing a smaller observed band after changing the instrument or model does not constitute convergence to exactness. Independent channels and transport across resolution are more informative than repeated analyses of one kernel.

Symmetric error-control budget

The confirmatory multiplicity budget is symmetric: it controls false favorable support across representation, origin, routing, endpoint, missingness, route, rival, tolerance, and subgroup searches, and false counterexample declaration across streams, endpoints, instruments, and models. Adaptive choices remain exploratory or inside training. A protocol-fixed deterministic falsifier, once verified on held-out data with its truth bridge intact, is not weakened by unrelated multiplicity; a noisy search for such a falsifier remains subject to multiplicity control. Promotion and retirement each require their protocol-fixed outer decision set.

Sequential evidence across repeated cohorts

Repeated inspection requires a rule valid over the inspection times. Let E_n be nonnegative and adapted to the recorded data filtration, with $E_0 = 1$, and a supermartingale under every law in the declared null family $\mathcal{H}_0$. For $0 < \alpha < 1$, Ville's inequality gives

$$\sup_{Q \in \mathcal{H}_0} Q\left(\sup_{n \geq 0} E_n \geq \tfrac{1}{\alpha}\right) \leq \alpha.$$

Stopping at the first crossing, applying the supermartingale expectation bound at each finite horizon, and increasing the horizon proves the statement. This supports valid repeated looks under the stated null, unlike repeatedly applying a fixed-sample threshold. Confidence sequences provide the corresponding time-uniform estimation approach (Howard et al., 2021).

For a concrete one-sided choice test, let Y_n indicate a continuing-life outlet choice among the prospectively defined recorded choosers. If the conditional null gives $\mathbb{E}(Y_n \mid \mathcal{F}_{n-1}) \leq q_n$, with known predictable $0 < q_n < 1$, choose a predictable $0 \leq \lambda_n \leq 1/q_n$ and use

$$E_n = E_{n-1}\{1 + \lambda_n(Y_n - q_n)\}.$$

The factor is nonnegative and has conditional expectation at most one. In the independent concealed-mapping design of Section 14, $q_n = 1/20$ only when the null independence includes choice recording and the actual allocation rule. Blocking, leakage, or conditioning that changes that probability

must be modeled rather than ignored. Dependence between animals is allowed only insofar as the conditional bound remains valid; otherwise use a justified cage- or batch-level construction. Allocate error across the prespecified history, duration, timing, and developmental families. Products of evidence from dependent stages do not acquire validity merely by being called cumulative evidence. A crossing tests the declared null, not every premise of PTVC.

17. Falsification, safeguards, ethics, and research agenda

Results that would narrow or reject a protocol-fixed PTVC claim

Because the program is built to learn, clear counterevidence would be a valuable scientific result. A deterministic, nonempty truth-containing residual set for a gate-valid stream that excludes zero refutes the protocol-fixed exact set-universal class. Other branch-matched results include endpoint independence with demonstrably nonzero ledger occupation; imbalance outside validated remaining capacity; reproducible scalar or bipolar representation failure; origin flexibility that manufactures closure; cross-channel incompatibility; and repeated route failure against adequate equal-budget rivals. Frequentist adverse decisions preserve coverage and multiplicity; posterior adversity remains prior- and model-conditional. A better-fitting no-atom model weakens the exact-atom branch; memory-only closure favors memory rivals; missing-not-at-random or endpoint sensitivity can remove terminal support; and adequate-rival parity removes differential route support.

Known challenges and their research solutions

Asymmetric outcome adjudication. A preregistered null result at adequate precision counts against the specific history-by-choice or post-commitment dependence extension that predicted enrichment, but does not by itself refute the formal zero-residual target. A positive pre-choice result with failed cue controls warrants apparatus redesign; a positive post-commitment result

with verified independent timing warrants an adversarial audit of randomizer records, logging order and possible dependence. Convergent history, feeding-delay and developmental effects gain evidential force only when each component survives its own controls and shared-cohort dependence is propagated. A separately validated complete-stream residual set excluding zero is a different and potentially target-adverse result. Publishing these distinctions before outcome access prevents a null from being rescued by moving the target and a positive result from being promoted beyond its evidence.

Interpretation of experimental outcomes

A choice rate consistent with 1/N under a well-powered test indicates that no enrichment was detected at the registered precision. It updates the specified route model but does not alone refute the universal terminal-zero target.

Above-null pre-choice matching with failed cue or mapping audits remains an unresolved apparatus or information-pathway finding. The apparatus must be revised and the result repeated before any PTVC-specific mechanism inference is warranted.

Replicated pre-choice enrichment with successful controls and a frozen history signature provides evidence against the registered local-cue comparator and favors the tested selector model. It does not by itself identify a phenomenal-valence ledger.

Replicated post-commitment dependence between choice C and safe designation K, with independently audited later

randomization, is evidence against the stipulated joint-independence model. Nonlocal, global, and methodological explanations would require examination. This result does not prove that the safe outlet was identifiable before commitment or uniquely establish PTVC.

A pre-treatment hatching effect associated with future assignment requires separate scrutiny of allocation, batch effects, handling, and the whole-trajectory extension. It does not automatically establish retrocausation or complete-stream closure.

A valid complete-stream residual object excluding zero provides potential branch-appropriate adverse evidence for [SET-ALL], subject to its truth guarantee. This result does not depend on success of the mosquito selector.

Phenomenal sign, cardinal or interval scale, neutral origin, exposure, stream individuation, ownership, and truth-level endpoint are demanding latent constructs. Complete histories are rarely observed, endpoint and observation processes are entwined, and some models may remain partially identified. The architecture turns these challenges into explicit measurement, partial-identification, transport, and decision problems, allowing each result to retain a clear representation, bearer class, species, culture, endpoint cause, and resolution.

Ethical boundary

PTVC is a research hypothesis; clinical, legal, moral, and policy decisions retain their independent evidentiary and ethical

foundations. Human research follows applicable requirements, preserves informed consent and withdrawal, protects privacy, minimizes burden, and receives independent oversight (World Medical Association, 2024). Animal work follows applicable requirements and replacement, reduction, and refinement principles, with transparent reporting (Percie du Sert et al., 2020). This framework permits prospectively reviewed empirical terminal experiments in mosquitoes, including the one-continuation versus nineteen-terminal-outlet design in Section 14. Approval concerns the actual species, conditions, numbers, endpoint procedure, containment, and scientific justification; it is not supplied by assuming that PTVC is true. Simulation remains a design and analysis tool rather than an exclusive setting for terminal tests.

What should materially change belief

A strong favorable pattern would combine a frozen valence representation, stable origin and exposure rule, method-diverse channels, one joint truth-containing prefix–observed–suffix object, a validated endpoint and observation model, contraction of the latent compatibility set under linked resolutions, one completion potential transported across modules, superior held-out performance against adequate equal-budget rivals, correct endpoint-surprise behavior, and independent replication. Such a result would constitute strong support for the protocol-fixed class, branch, and resolution while defining the next quantifier-bridge challenge.

These criteria make belief revision constructive: favorable signatures gain force as protocol-fixed bridges pass, while resolved ambiguities narrow the live explanations. Near-zero means, adaptation, terminal decline, cultural recurrence, fitted bridges, and partition slopes gain evidential weight through independent measurement, transport, and equal-budget rival comparison; nonsignificance alone remains nonadjudicative.

Near-term prerequisite and differential predictions

One near-term check is Section 9's random-cut restriction: $\beta = 1$ (prefix slope -1) under the declared representation and an identifying or bounding error model. Agreement supports consistency with the additive partition, not its full phenomenal validity or PTVC closure. A nonzero intercept supports nonclosure only when its error location and complete-stream domain are identified or validly bounded; for a bounded episode, the conclusion remains local. A failed slope directs review to representation, sampling, or the observation bridge.

Early publishable milestones

A valuable paper can materially advance PTVC by validating one frozen representation, origin rule, exposure clock, observation kernel, missingness model, endpoint interval, or local route in a protocol-fixed class. It should report generator recovery, control performance, uncertainty, and transport. Its immediate contribution is a robust measurement or local mechanism result and a rigorous foundation for terminal studies.

Cumulative decision bookkeeping

Each result enters a versioned evidence ledger recording the target and modality, bearer and routing, representation, origin and exposure, endpoint, observation and missingness, reference law and support, tolerance or atom claim, rivals and budget, estimand and proper score, recovery, decision, and claim ceiling. Outcomes downstream of one perturbation remain one evidence family. Promotion requires its held-out gate; no index offsets a failed target or truth bridge. Replication changes a principal bias (site, instrument family, analyst team, endpoint process, species, or intervention) while transporting the same frozen object.

Promotion, retirement, and open challenge

The project should maintain a public generator bank that includes supportive, rival, mixed, singular, near-zero, measurement-failure, missing-not-at-random, and none-of-the-above worlds. New routes enter with independent equations and prospective retirement rules. A route retires after reproducible recovery failure, failure of its defining equation under adequate power, or loss to an equally resourced rival across its declared support. A target branch retires when a gate-valid counterexample satisfies its protocol-fixed falsifier. Research involving vulnerable human or endpoint-proximal clinical populations preserves survival and care and receives the additional review appropriate to that population. The mosquito branch follows a separately specified empirical design submitted for independent scientific, welfare, and containment review

before implementation. The same infrastructure makes adverse, ambiguous, and no-adequate-estimate outcomes as publishable as favorable ones.

Ordered research agenda

The agenda combines independent formal review and generator-recovery challenges with representation, observation, and bounded-route studies. Random-cut, signed-dose, horizon, and completion-potential work can proceed alongside the empirical mosquito terminal-choice, duration, post-event-delay, and future-regime studies in Section 14. Each module has its own prospectively fixed prediction and appropriate validation; simulations help assess recovery and error control without replacing the biological experiment. Animal testing is not a prerequisite for unrelated human or theoretical branches. Endpoint-aligned cohort analyses, low-burden multimethod studies, and direct complete-stream tests follow the measurement and endpoint evidence each requires. Each stage records outcomes, versioned protocols, reproducibility materials, and predetermined promotion and retirement rules.

A joint uncertainty model is particularly important in the mosquito study: variation in hatching, adult emergence, feeding, choice, and endpoint observation changes the analyzed population at successive stages. Separate the assignment-based effect from a survivor-only contrast, and carry dependence across these stages rather than treating each favorable endpoint as a fresh independent test. The experiment adds an action-

selection and developmental extension to PTVC; it does not alter the signed-measure target, the strict-pre-endpoint convention, or the distinction between route and target evidence.

Program-level conclusion. PTVC asks whether the complete history of conscious experience obeys a precise boundary condition at its true endpoint. This possibility has consequences that mathematics can sharpen and nature can test: the signed contributions within a qualifying stream may satisfy an exact intrastream equality, with distinctive pressure on endpoint timing, reachability, retained information and choice. The random-endpoint proposition identifies a sharp structural restriction on independently timed endings. The randomized-outlet program adds an exact behavioral null for investigating one proposed whole-trajectory extension. Together with calibrated measurement, strong rival models and open replication, these ideas create a substantive research opportunity across consciousness science, affective measurement and stochastic dynamics. A reproducible, prospectively predicted signature would invite a new generation of experiments into a potentially unrecognized organizing principle of conscious life.

18. Evaluation and reproducibility standard

Evaluation asks whether the formal target and research architecture meet internal-validity, claim-ceiling, comparative-testability, ethical-feasibility, and reproducibility standards. Meeting them makes PTVC scientifically assessable and creates a disciplined path for empirical support to accumulate through prospectively specified studies.

Modular, noncompensatory evaluation

Evaluation keeps five modules separate: formal coherence; measurement and identification; endpoint and ownership validity; recovery and error control; and ethics and reporting. Strengths and limitations therefore remain local to the affected branch, assumption, estimand, or verdict. A limitation narrows or retires that branch; a blocked complete-stream object receives NAE until its bridge is supplied.

Minimum reproducibility capsule

Every empirical study derived from this framework should archive a timestamped protocol; claim, formula, bearer, routing, endpoint, observation, missingness, generator, data-split, software, seed, and code records. Data provenance, exclusions, transformations, linkage, access restrictions, and time windows must be documented. Outputs should include machine-readable decisions, recovery matrices, acceptance geometry, planned sensitivities, and public-artifact hashes. Publish executable synthetic fixtures, schemas, code, and decision records where lawful and safe; use governed access or documented substitutes

for restricted materials and state which independent checks remain possible.

Independent audit and discrepancy handling

Before outcome access, an independent audit checks object typing, formula order, endpoint and cut randomization, no-lookahead, action domains, policy mass, rank, weighting, multiplicity, and open-set behavior. Corrections are versioned without replacing the original protocol, output, or verdict. Distributed formats and machine-readable sources must match the displayed mathematics and claims. Production fixes leave the scientific model unchanged; scientific changes receive a new version.

Promotion, retirement, and publication outcomes

Publication outcomes follow Sections 14 and 17’s protocol-fixed promotion and retirement rules. Every protocol-defined outcome remains publishable; verified counterexamples remain attached to the tested version, finite-resolution compatibility remains resolution-limited, and no target-defining element may move after inspection.

Declarations

Author: Stephen Takowsky

Funding. No external funding.

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Competing interests. Stephen Takowsky originated PTVC, founded the Law of Fairness Project, and owns or controls rights in related publications and project materials.

Data, code, and materials. The operative definitions, propositions, and decision rules used here are stated in this article with their cited mathematical and methodological sources. No new participant dataset is reported. Studies derived from the framework should archive and share code, synthetic fixtures, schemas, and decision records under their prospectively specified ethics, privacy, licensing, and version-control plans.

Ethics statement. This theoretical article reports no completed human or animal experiment and uses no participant data. Section 14 sets out separate pre-choice and post-commitment mosquito proposals for prospective scientific, welfare, and containment review; no approval, preregistration identifier, completed animal trial, or experimental outcome is claimed.

Correspondence. team@lawoffairness.com.

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About the Author

Stephen Takowsky has spent his life studying complex systems: how rules shape them, how people cooperate and compete within them, and how hidden constraints can change what becomes possible. His path has crossed finance, competitive strategy, professional esports, and the leadership of global online communities, ultimately leading to the Law of Fairness project, a decades-long effort to investigate whether fairness itself may reflect a deeper structure of conscious life.

Competing as “MrGameTheory,” Takowsky reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues while pursuing a career in finance at Morgan Stanley in Beverly Hills. In 2016, aXiomatic, an ownership group led by Peter Guber and Ted Leonsis and backed by investors including Magic Johnson, Steve Case, and Tony Robbins, acquired a controlling interest in Team Liquid. Following the acquisition, Team Liquid selected Takowsky to captain its newly created Civilization team, later recognized by Guinness World Records as the first Civilization esports team. Across thousands of games, Civilization became an unconventional education in strategy: understand the whole system, think far beyond the next move, anticipate intelligent opposition, and keep searching when the obvious paths fail.

Strategy then became community leadership. Takowsky spent seven years at 505 Games, serving as Senior Digital Content & Engagement Manager and leading global communities for Terraria, which has surpassed 70 million copies sold, as well as Ghostrunner, WUCHANG: Fallen Feathers, Eiyuden Chronicle: Hundred Heroes, and other titles. He also founded the United Servers of Discord Alliance, bringing together leaders from major communities built around Roblox, Pokémon, Star Wars, League of Legends, Grand Theft Auto, Warframe, Animal Crossing, and other properties. Earlier, he served as administrator and producer for TF2Outpost, a major Steam digital-goods marketplace. Across these very different environments, he kept returning

to the same insight: better structure can expand what people are capable of building together.

Throughout those years, one question kept following him: could the apparent unfairness of life conceal a deeper structure? That question became the Law of Fairness. Takowsky began developing the idea more than twenty years ago while studying Religious Studies and Philosophy/Law and Society at the University of California, Riverside, and gave it its first public form in his 2009 book, *Of Grandeur*. In the years that followed, he sought out academics and serious thinkers across many disciplines, asking for insights, criticism, and sources that could help separate fact from speculation. What began as a philosophical intuition gradually became a research program built to face mathematics, measurement, competing explanations, and evidence.

The ten-book Law of Fairness series follows that question outward, from consciousness and scientific testing into ordinary life, meaning, and civilization. If the Law is true, then there may be no “free happiness” in the sense of positive experience remaining unresolved by the lifetime balance the hypothesis predicts. That possibility raises a profound practical question: what if humanity has been pursuing the wrong model of utopia? Rather than organizing life around the endless pursuit of pleasure, the series explores whether greater self-moderation, less needless suffering, and better-designed institutions could help remove some of the forces that keep people and societies trapped in conflict, corruption, and instability.

Takowsky’s fifty-year horizon is designed so that building a better future does not require tearing apart the present. It gives institutions and communities time to test reforms, build infrastructure, prepare future leaders, improve technology, and correct mistakes before they spread. Just as importantly, it gives today’s generation the chance to be remembered for beginning something larger than itself: laying the

foundations for a more transparent, peaceful, and prosperous world it may never live to see.

The project's future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers willing to bring rigor, imagination, and courage to one of humanity's oldest questions. The next breakthrough may come from a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. *The Law of Fairness* began as one person's question. Its next chapter belongs to everyone willing to help discover what is true and build what that knowledge makes possible.

Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.

